The Real Cost of Benchmarking*

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Abstract

This paper provides causal evidence that benchmarking-induced asset price distortions have real effects on corporate investment. We document that increased benchmarking over the past 20 years fundamentally changed the cross-section of stocks' CAPM $\widehat{\beta}$s. We establish causality using exogenous variation in stocks' benchmarking intensity around Russell index reconstitutions. Stocks' CAPM $\widehat{\beta}$s increase upon benchmark inclusion with larger effects for stocks which experience larger benchmarking intensity increases. Firm managers perceive this as an increase in their cost of capital and reduce investment. Treated firms have 7.1% less physical and 8.4% less intangible capital after six years. We find consistent results at the industry level using long-differences from 2000 to 2016. At the aggregate level, the increase in CAPM $\widehat{\beta}$s caused by benchmarking largely offset the decline in the risk-free rate over the past 20 years and can explain 57% of the missing investment puzzle.

JEL classification: D2, E22, G11, G23, G31

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1 Introduction

Over the past 20 years the U.S. economy has been shaped by two trends: weak corporate investment relative to valuations and a rise in benchmark-linked capital allocation. The growth of passive index funds and the evaluation of active funds against benchmarks means that a large share of capital today is allocated based on stocks' membership in benchmark indices, as opposed to fundamentals.[1] This inelastic demand distorts asset prices, leading to higher prices (Shleifer, 1986), increased volatility (Ben-David, Franzoni, and Moussawi, 2018), and greater co-movement (Barberis, Shleifer, and Wurgler, 2005) for stocks in benchmark indices. Whether these asset price distortions have contributed to weak corporate investment is not well understood.

This paper studies the causal effects of benchmarking-induced asset price distortions on corporate investment. We document a novel mechanism through which benchmarking influences corporate behavior. We use exogenous variation in stocks' benchmarking intensity to show that increased exposure to benchmark-linked capital flows raises stocks' CAPM β estimates (i.e., $\widehat{\beta}$). Firm managers interpret this increase in CAPM $\widehat{\beta}$ as a higher cost of capital and consequently reduce investment. Importantly, we show that these results are not driven by changing firm fundamentals. Instead, we argue that firm managers rely on textbook guidance to set discount rates using CAPM $\widehat{\beta}$s without accounting for distortions created by benchmarking. These distortions have a substantial impact on investment decisions at the firm, industry, and aggregate level through their effect on the perceived cost of equity capital. Our study provides new insights into how the growing trend of benchmark-linked investing affects real economic outcomes.

We illustrate our proposed mechanism in a stylized model that introduces two frictions into a standard model of corporate investment. The first source of friction are benchmarking-induced asset price distortions that drive wedges into firm discount rates (Kashyap, Kovrijnykh, Li, and Pavlova, 2021). The inelastic demand of benchmarked funds for benchmark constituent stocks raises their price, but also increases their co-movement. These forces have opposing effects on the discount rate: the increased stock price lowers the implied discount rate and incentivizes investment, while greater co-movement discourages it. As such, the overall effect of benchmarking on discount rates and optimal investment is ambiguous. The second friction is a behavioral assumption that firm managers behave exactly as they are taught to in textbooks and MBA classrooms: they use the weighted average cost of capital implied by the CAPM to discount future cash flows.

[1] In 2023, $17.9 trillion in assets were benchmarked to S&P Dow Jones' and $10.5 trillion to FTSE-Russell's U.S. indices.

While the assumption that firm managers practice what textbooks teach[2] may seem innocuous, it is key to our mechanism. Firm managers who set discount rates using stocks' CAPM $\widehat{\beta}$ will observe an increase in co-movement upon benchmark inclusion which discourages investment. However, they will overlook the price effect that incentivizes investment. This failure to internalize the distortionary effects of benchmarking leads managers to perceive an increase in their cost of capital. Consequently, benchmarking has an unambiguously negative impact on investment.

We test our model's predictions using the benchmarking-intensity measure (BMI) developed by Pavlova and Sikorskaya (2023). BMI measures the total inelastic demand that a stock attracts from benchmarked funds, expressed as a fraction of the stock's market capitalization. We merge the BMI measure with CAPM $\widehat{\beta}$s from Welch (2022b), accounting data from Compustat, data on managers' perceived cost of capital from Gormsen and Huber (2024), and stock market data from CRSP from 1998 to 2018.

We begin by documenting new facts about the cross-section of CAPM $\widehat{\beta}$s. We show that over the past 20 years, stocks' CAPM $\widehat{\beta}$s and benchmarking intensity increased in lockstep. Firms representing over 40% of annual capital expenditures in Compustat experienced a significant increase in their CAPM $\widehat{\beta}$, with an average increase of 0.33. However, this increase in CAPM $\widehat{\beta}$s is not due to changes in firms' fundamental risk, as measured by cash flow βs, or changes in leverage. Instead, we find distortions in the cross-section of CAPM $\widehat{\beta}$s across market equity ranks used to construct benchmark indices. This suggests that the rise in benchmark-linked capital allocation affected the measurement of CAPM βs over the past 20 years.

We establish a causal link between benchmarking and CAPM $\widehat{\beta}$ using a difference-in-differences design around Russell index reconstitutions. The Russell indices are widely used benchmarks for U.S. equity markets and are reconstituted annually based on market capitalization ranks. Changes in Russell index membership around benchmark inclusion cutoffs lead to exogenous changes in benchmarking intensity. Our difference-in-differences approach does not require that benchmark inclusion is random or common support in covariate levels across stocks. It only requires that treated and control stock's CAPM $\widehat{\beta}$ would have evolved similarly absent changes in benchmarking intensity. We restrict our sample to all stocks within 250 ranks around the Russell index cutoffs to ensure we capture only changes in benchmarking intensity due to index reconstitution. We control for stocks' momentum over the past 12 to 24 months before index reconstitution to account for the possibility that stocks' momentum affects CAPM $\widehat{\beta}$. We further include

[2]For example, corporate finance textbooks by Brealey, Myers, Allen, and Edmans (2023), Berk and DeMarzo (2023), and Ross, Westerfield, Jaffe, and Jordan (2016). A notable exception is Welch (2022a, Chapter 10) who suggests to always use a market β of 1 to calculate discount rates. Gollier (2021) estimates that the welfare loss from using a single discount rate is equivalent to a permanent reduction in consumption of up to 45% in a calibrated Lucas model.

high-dimensional fixed effects that remove as much time-varying unobserved heterogeneity as possible to ensure that our estimates are well-identified.

The difference-in-differences results show that an increase in the BMI of a stock by at least 5 percentage points (p.p.) due to index reconstitution increases its CAPM $\widehat{\beta}$ by 0.18. The treatment effects of benchmark inclusion on CAPM $\widehat{\beta}$ increase with benchmarking intensity: Stocks with a BMI increase of at least 10 p.p. (20 p.p.) subsequently have 0.23 (0.35) higher CAPM $\widehat{\beta}$s. Micro-cap stocks do not drive the effects. The effects are also present for stocks with market capitalizations above the 20th percentile of the New York Stock Exchange (NYSE). The changes in CAPM $\widehat{\beta}$s are not due to changes in fundamentals or increased risk exposure. Instead, our results show that increased co-movement with the market is explained by institutional ownership and exposure to benchmark-linked capital flows, similar to the findings of Antón and Polk (2014).

The increase in firms' CAPM implied cost of equity is substantial and persistent. Assuming an annual equity risk premium (ERP) of 6%, our baseline results imply an increase of 108 basis points (bps). The effect persist for years and only partly reverses: we find a 125 bps higher cost of equity after one year, 75 bps after four years, and 67 bps after seven years. If managers rely on the CAPM to allocate capital, persistent changes in $\widehat{\beta}$s have long-lasting effects on investment behavior. Compared to other discount rate shocks these effects are fast, large, and persistent.[3]

In contrast, effects on the implied cost of capital (ICC) derived from stock prices are short-lived. We find a 36 bps decrease in the ICC at benchmark inclusion[4] fading to 14 bps after one year and becoming insignificant thereafter. Following Eskildsen, Ibert, Jensen, and Pedersen (2024) we calculate the ICC by averaging four popular models from the accounting literature: the residual income models from Gebhardt, Lee, and Swaminathan (2001) and Claus and Thomas (2001), and the dividend discount models from Easton (2004) and Ohlson and Juettner-Nauroth (2005). Our findings suggest that benchmarking distortions primarily affect the cost of capital through persistent changes in CAPM $\widehat{\beta}$s, rather than short-term fluctuations in the stock price.

We document that increases in a stock's benchmarking intensity predict a higher perceived cost of capital by the firm's managers as reported in the data collected by Gormsen and Huber (2023). We use changes in BMI as an instrumental variable (IV) to identify the causal effect of changes in CAPM $\widehat{\beta}$ on a firm's perceived cost of capital. The estimates show that managers' perceived cost of capital responds to benchmarking-induced increases in CAPM $\widehat{\beta}$: a 0.20 change in CAPM $\widehat{\beta}$ causes an increase in managers' perceived cost of capital of approximately 74 bps.

Our second set of results estimates how firms react to changes in their CAPM β induced by

[3]For example, Bauer and Rudebusch (2020) estimate that the natural rate declined by 100 bps between 2002 and 2020.
[4]A 36 bps decrease implies a 5.24% price increase, close to the 5% documented by Chang, Hong, and Liskovich (2015).

exogenous changes in their stock's benchmarking intensity. For a firm manager who follows textbook guidance to set investment policies using the WACC implied by the CAPM, an increase in CAPM $\widehat{\beta}$ raises the user cost of capital and should lead to a decline in investment.

We test whether changes in CAPM $\widehat{\beta}$ affect firm outcomes like capital expenditure, physical and intangible capital stocks, cash holdings, payouts, and employment. We use Jordà's (2005) local projections (LP) to estimate the effects of an increase in CAPM $\widehat{\beta}$ on capital allocation over horizons of up to 6 years using changes in BMI as IV. The instrument uses the plausibly exogenous variation in BMI from Russell index reconstitution to instrument for the endogenous relationship between CAPM β and investment. We ensure that our estimates are well-identified using three methods. First, we saturate our LP-IV estimator with high-dimensional fixed effects to remove as much time-varying unobserved heterogeneity as possible. Second, we confirm that our results are robust to inclusion of known predictors of capital accumulation (e.g., Tobin's Q or cash flow). Third, we conduct several tests to validate the exclusion restriction but find no evidence that changes in BMI correlate with changes in risk exposure, debt market access, or corporate governance: The CAPM $\widehat{\beta}$ of peer firms remains stable when a treated firm's BMI changes, and firm-level risk measures show no correlation with changes in BMI. BMI changes do not affect measures of financial frictions and the cost of debt, including CDS spreads. Corporate governance scores do not change when BMI changes. These findings suggest that BMI changes are orthogonal to other factors that influence investment and support their use as valid instrument.

A benchmarking-induced increase in a firm's CAPM $\widehat{\beta}$ leads to a substantial and persistent reduction in investment. Specifically, a 20% rise in CAPM $\widehat{\beta}$ results in a cumulative decrease of 10.0% in capital expenditure over a six-year period. Instead of investing, treated firms first accumulate cash and subsequently increase payouts to shareholders. The treatment effects are consistent with firms using rolling window estimators that gradually update their CAPM $\widehat{\beta}$ estimates: they are near zero for short horizons, grow over time, and become statistically significant after three years. Over six years the average treated firm's physical capital stock declines by 7.1%; its' intangible capital stock declines by 8.4%. The estimated response of capital stocks imply a user cost of capital elasticity close to the theoretical value of 1 implied by Cobb-Douglas production.

We find supporting evidence in the NBER-CES manufacturing data showing that higher CAPM $\widehat{\beta}$s lowered industry-level capital accumulation by 12.5% from 2000 to 2016. The dataset includes over 100 industries, covering both public and private firms.[5] To estimate the impact of rising industry-level CAPM $\widehat{\beta}$s on capital accumulation, we use IV regressions in long-differences

[5]Private firms also use the CAPM to set discount rates for investment decisions. Since their equity is not publicly traded, they often rely on comparable public firms to estimate their cost of equity.

from 2000 to 2016. We calculate the value-weighted change in CAPM $\widehat{\beta}$ and instrument it with the value-weighted change in BMI at the industry level. The results are robust to the inclusion of variables that capture pre-trends at the industry level as well as to inclusion of sectoral fixed effects that force the identification to come from within-sector variation.

Lastly, we calculate a counterfactual weighted average cost of capital (WACC) for the average firm in our sample by adjusting the CAPM $\widehat{\beta}$ for benchmarking-induced distortions. This reveals an average wedge of 145 bps between the actual and counterfactual WACC since 2004. Over the past 20 years, the equal-weighted average CAPM $\widehat{\beta}$ increased by 0.41 after remaining relatively stable from 1975 to 2000. We attribute 88% of this increase to the 14 percentage point rise in average benchmarking intensity from the early 2000s. We adjust the average CAPM $\widehat{\beta}$ for this increase and estimate a counterfactual WACC. Our findings show that the decline in the risk-free rate over the past two decades has been largely offset by the increase in CAPM $\widehat{\beta}$s.

We assess whether the wedge between actual and counterfactual WACC is sufficient to explain the puzzle of missing aggregate investment documented by Gutiérrez and Philippon (2017). Using 1990–2002 data, we estimate the historical relationship between aggregate investment and Tobin's Q, then predict post-2002 investment assuming this relationship remains constant. The cumulative shortfall since 2002 represents the "missing investment". We adjust Tobin's Q following Gormsen and Huber (2023) to account for the discrepancy between the market's discount rate and firms' perceived cost of capital.

The WACC wedge we document can explain 57% of the missing investment puzzle at the aggregate level. Without adjustment, the investment shortfall implied by Tobin's Q is approximately 25% of the capital stock by 2019. After accounting for the WACC wedge created by benchmarking, this shortfall falls to about 11%. The remaining gap likely stems from other factors, such as rising market power (Barkai, 2020, Crouzet and Eberly, 2023).

The paper is organized as follows: The remainder of this section discusses related literature. Section 2 documents several new facts about the cross-section of CAPM $\widehat{\beta}$s. Section 3 illustrates our proposed mechanism in a stylized model. Section 4 describes the data. Section 5 establishes a causal link between benchmarking and CAPM $\widehat{\beta}$. Section 6 documents that benchmarking-induced changes in CAPM $\widehat{\beta}$ affect real outcomes at the firm- and industry-level. Section 7 studies whether the CAPM $\widehat{\beta}$ distortions caused by benchmarking are large enough to explain the missing investment puzzle at the aggregate-level. Section 8 concludes.

Related Literature This paper contributes to several strands of literature, including the effects of benchmark-linked investing on asset prices, corporate behavior, and capital (mis-)allocation.

The literature on benchmark-linked investing, starting with Shleifer (1986) and Harris and

Gurel (1986), established that stocks appreciate when included in an index and that stock volatility (Ben-David et al., 2018) and co-movement with the index (e.g, Barberis et al., 2005, Boyer, 2011) increase after benchmark inclusion (see Wurgler, 2010, for a survey). We provide causal evidence that a stock's benchmarking intensity determines its CAPM $\widehat{\beta}$ and document that the CAPM $\widehat{\beta}$s increased in lockstep with benchmarking intensity over the past 20 years. We show that these increases in CAPM $\widehat{\beta}$s are not due to changes in firm fundamentals, leverage, or risk exposure as measured by cash flow βs. Rather, benchmark-linked capital flows drive a wedge between measured CAPM βs and cash flow βs that is unrelated to the risk exposure of firms' cash flows.

Basak and Pavlova (2013) provide a theoretical framework showing that performance benchmarking of asset managers leads to asset class effects consistent with the observed benchmark inclusion effects.[6] Building on this, Kashyap et al. (2021) derive optimal corporate investment policies in the presence of benchmarked funds, arguing that firm managers should internalize the inelastic demand for benchmark stocks and invest more to maximize firm value. We contribute to the discussion about the effects of benchmarking on real investment in two ways: empirically, we provide causal evidence that increases in benchmarking lead to lower investment and a higher perceived cost of capital. Conceptually, we introduce a behavioral argument that reconciles our findings with Kashyap et al. (2021). We argue that firm managers, relying on textbook guidance to estimate the cost of capital using the CAPM, fail to internalize benchmarking-induced asset price distortions. This results in an overestimation of the cost of capital and a sub-optimal decline in investment, ultimately destroying shareholder value.

Additionally, our paper contributes to the literature on how firm managers set discount rates. Despite the CAPM's failure to explain the cross-section of expected stock returns (Fama and French, 2004),[7] it reigns supreme in practice: Welch (2008) reports that about 75% of finance professors recommend using the CAPM and the Duke CFO survey (Graham, 2022) finds that the CAPM is the leading method to determine discount rates. Further evidence from earnings calls (Gormsen and Huber, 2024), M&A transactions (Dessaint, Olivier, Otto, and Thesmar, 2020), and mutual funds (Berk and Van Binsbergen, 2016) confirms that practitioners rely on the CAPM. We add by documenting that benchmarking induced CAPM distortions have first-order effects on the discount rates that managers use to make capital budgeting decisions. We argue that these distortions have become large enough over the past 20 years to affect the economy as a whole.

[6] Kim (2024) shows that benchmarking can lead to procyclical risk-taking by asset managers which creates correlated demand shocks that amplify stocks' market risk. See also Cuoco and Kaniel (2011), Buffa, Vayanos, and Woolley (2022), and Buffa and Hodor (2023).

[7] Berk, Green, and Naik (1999) show that stock returns need not satisfy the CAPM even when expected returns on all individual projects do, since a firm's stock also embeds real options to undertake new and abandon old projects.

Despite the widespread use of the CAPM in practice, the effects of variation in CAPM $\widehat{\beta}$s on investment are not widely studied. The literature primarily focuses on how the cost of debt (Gilchrist and Zakrajšek, 2007, Philippon, 2009) or tax policy (Zwick and Mahon, 2017, Mark, Garrett, Ohrn, Roberts, and Suárez Serrato, 2021, Matray, 2023) affect investment. Notable exceptions are Krüger, Landier, and Thesmar (2015) and Frank and Shen (2016). Krüger et al. (2015) document investment distortions caused by the use of a single discount rate within firms. Using OLS regressions, Frank and Shen (2016) find that higher contemporaneous CAPM $\widehat{\beta}$s are associated with higher investment. In contrast, we use an IV approach to address the endogenous relationship between CAPM βs and investment (Berk et al., 1999, Zhang, 2005, Kuehn and Schmid, 2014) and forecasts the effects of CAPM distortions up to six years into the future. Consistent with theoretical predictions, we find that exogenous increases in CAPM $\widehat{\beta}$s lead to lower investment.

Our findings also link to a literature on the corporate governance implications of passive investing. Appel, Gormley, and Keim (2016), Edmans and Holderness (2017), Bebchuk, Cohen, and Hirst (2017), Heath, Macciocchi, Michaely, and Ringgenberg (2021), and Lewellen and Lewellen (2022) examine the effects of the rise in passive investing on corporate governance. Our results show that benchmark-linked investing affects corporate investment without an explicit corporate governance channel through its impact on the perceived cost of equity capital.

2 New Facts About CAPM β Estimates and Benchmarking

This section presents new facts about CAPM $\widehat{\beta}$s in the cross-section of U.S. stocks. We document that over the past 20 years stocks' CAPM $\widehat{\beta}$s and benchmarking intensity have increased in lockstep. Firms representing over 40% of annual capital expenditures in Compustat experienced a significant increase in their CAPM $\widehat{\beta}$, with an average increase of 0.33. However, this increase in CAPM $\widehat{\beta}$s is not due to changes in firms' fundamental risk, as measured by cash flow βs, or changes in firms' leverage. Instead, we find distortions in the cross-section of CAPM $\widehat{\beta}$s across market equity ranks used to construct benchmark indices. This suggest that the rise in benchmarking has affected the measurement of CAPM βs, a possibility we investigate using causal inference techniques in Section 5.

We use monthly estimates of benchmarking intensity (BMI) from Pavlova and Sikorskaya (2023) from 1998 to 2018. BMI measures the amount of capital that inelastically demands the

equity of a publicly traded firm. The BMI for stock i in month t is defined as

$$\text{BMI}_{i,t} = \sum_{j=1}^{J} \frac{\lambda_{j,t} \times \omega_{i,j,t}}{\text{Market Equity Value}_{i,t}} \tag{1}$$

in which $\lambda_{j,t}$ are the assets under management (AUM) of mutual funds and ETFs benchmarked to index j and $\omega_{i,j,t}$ is stock i's weight in index j. We combine the BMI measure with estimates of stocks' CAPM βs. We use the estimator proposed by Welch (2022b) to calculate the market risk exposure of all common U.S. stocks with respect to the CRSP value-weighted index.[8] We split our sample into two periods: a pre-period from 1998 to 2002 and a post-period from 2003 to 2018.

We compare the distribution of conditional means of stocks' BMI and CAPM $\widehat{\beta}$ across market equity ranks. We fix a stock's market equity rank at the end of May and plot the stock's BMI and CAPM $\widehat{\beta}$s from June to May of the following year against that rank. This procedure mimics the construction of the Russell benchmark stock indices which are based on end of May equity ranks and reconstitute every year in June. For example, the Russell 3000, which comprises the largest 3000 stocks by market equity capitalization at the end of May. Pavlova and Sikorskaya (2023) estimate that 72.6% of the average stock's BMI is contributed by funds benchmarked to Russell indices. Beyond this, market equity ranks in May hold no inherent economic significance.

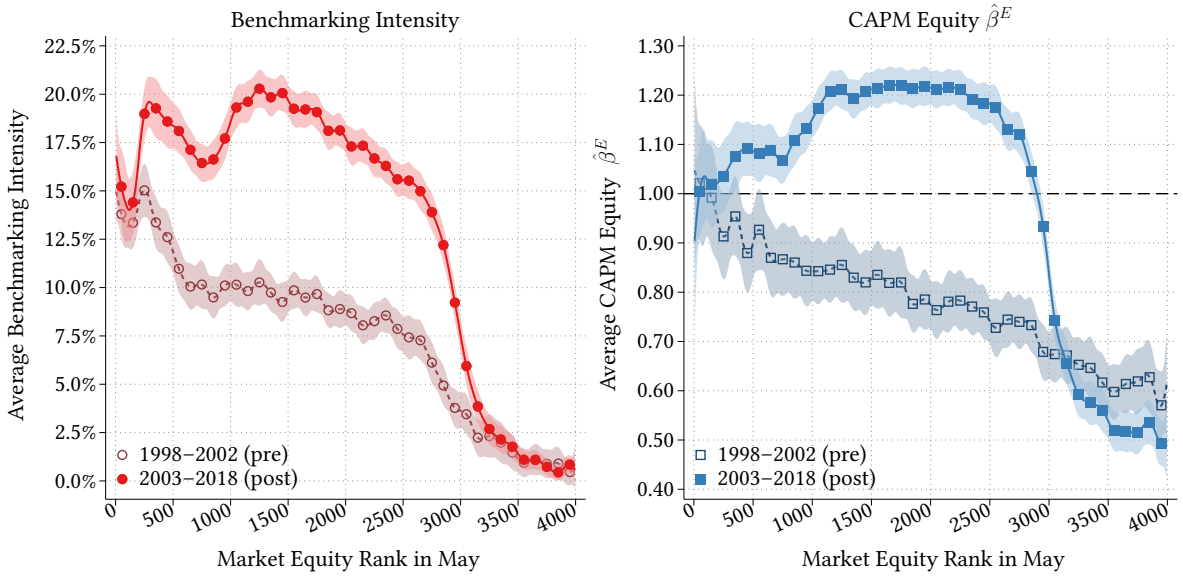
To analyze the raw data, we group stocks into bins of approximately 100 consecutive ranks.[9] For each bin, we calculate the equal-weighted mean of BMI and CAPM $\widehat{\beta}$. We use monthly data and absorb year-month fixed effects to identify the conditional means across market equity ranks using only the cross-sectional variation within each month. We then plot pooled estimates of the conditional means across market equity ranks for the pre- and post-periods using the non-parametric binscatter methods developed by Cattaneo, Crump, Farrell, and Feng (2024).

Fact 1: Stocks' benchmarking intensity correlates with CAPM $\widehat{\beta}$. Both stocks' average benchmarking intensity and CAPM $\widehat{\beta}$s have increased drastically since 2003. Figure 1 shows binned scatter plots of stocks' monthly BMI (left panel) and CAPM $\widehat{\beta}$ (right panel) against their market equity ranks. The average BMI has increased across all market equity ranks since 2003. Similarly, average CAPM $\widehat{\beta}$s have increased for almost all ranks. For stocks ranked between 1000th and 3000th (i.e., Russell 2000 stocks), benchmarking intensity more than doubled, with BMI rising from an average of 7.6% pre-2003 to 16.7% post-2003. Similarly, these stocks' average

[8] The estimator first winsorizes daily stock returns at -2x and +4x the contemporaneous market return. It then estimates β using WLS regression with exponentially decaying weights of 4-month half-life on an expanding window. Welch (2022b) shows that this estimator outperforms other estimators in predicting future CAPM $\widehat{\beta}$s out of sample.

[9] Finer partitions, e.g. bins of 40 ranks, yield similar results. See Online Appendix Figure B2.

Figure 1: Benchmarking Intensity and CAPM Equity $\widehat{\beta}^E$ versus Market Equity Rank in May



Notes: This figure shows binned scatter plots of stocks' monthly BMI and CAPM equity $\widehat{\beta}^E$ estimates vs. their market equity rank in May. Each bin shows the equal-weighted average of approximately 100 ranks. The conditional means across market equity ranks are identified using only cross-sectional variation by absorbing year-month fixed effects. Outlined bins show estimates from 1998 to 2002, while filled bins show estimates from 2003 to 2018. Shaded areas are 95% confidence bands based on standard errors clustered at the stock and year-month level.

CAPM $\widehat{\beta}$ increased by 53%, from 0.76 to 1.16.[10]

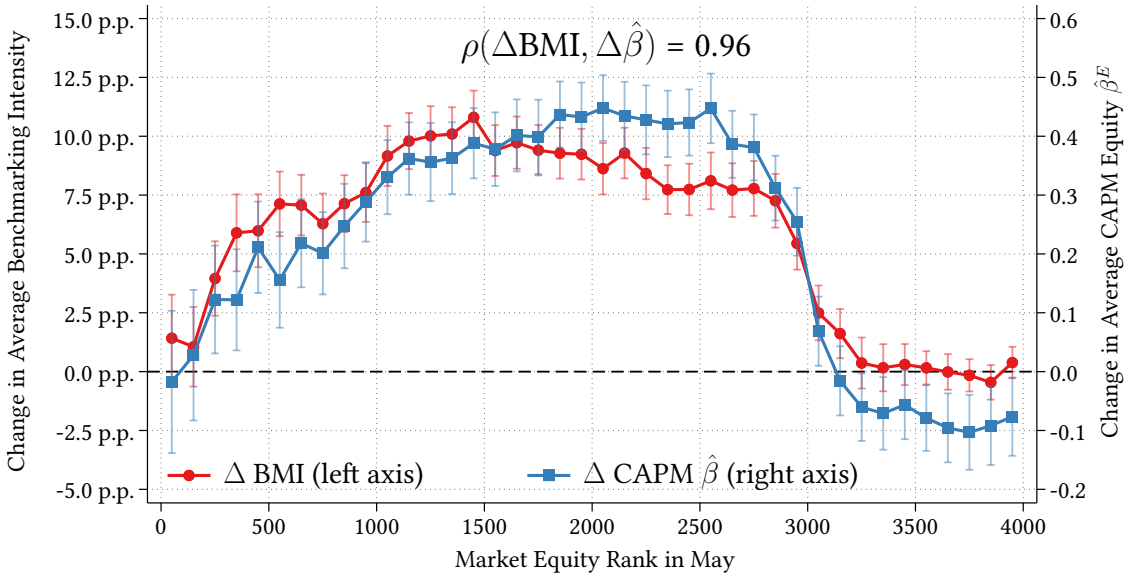
Changes in average benchmarking intensity and CAPM $\widehat{\beta}$ line up surprisingly well across market equity ranks. Figure 2 plots the difference in conditional means of BMI and CAPM $\widehat{\beta}$ estimates between the pre- and post-periods. Differences in BMI and CAPM $\widehat{\beta}$s across market equity ranks are highly correlated (ρ=0.96). Appendix Figure A1 shows that this strong correlation is also visible in the time series of BMI and CAPM $\widehat{\beta}$ levels from 1998 to 2018.[11]

The distribution of both BMI and CAPM $\widehat{\beta}$ estimates exhibit significant changes at equity ranks which are used as index cutoffs by Russell. In Figure 1, both BMI and CAPM $\widehat{\beta}$s increase around the 1000th rank, which is the cutoff for the Russell 1000/2000 indices. BMI and CAPM $\widehat{\beta}$s also decrease sharply around the 3000th rank, which is the cutoff for the Russell 2000/3000 indices. This suggests that the observed changes in CAPM $\widehat{\beta}$s are related to the design of benchmark indices. In Section 5, we exploit exogenous variation in BMI from Russell index reconstitutions

[10] Appendix Figure A2 shows similar increases in CAPM $\widehat{\beta}$s using daily, weekly, or monthly rolling-windows. Online Appendix Figure B1 confirms robustness to the Scholes-Williams (1977) adjustment for non-synchronous trading.

[11] An animated version of the time series can be found at https://www.christiankontz.com/files/Beta_BMI_over_time.gif

Figure 2: Differences in Benchmarking Intensity and CAPM Equity $\widehat{\beta}^E$ Between Pre and Post



Notes: This figure shows changes in average BMI (left axis) and CAPM equity $\widehat{\beta}^E$ (right axis) between the pre (1998–2002) and post (2003–2018) periods by market equity ranks in May. Each bin represents the difference in conditional mean estimates of BMI and CAPM $\widehat{\beta}$s from Figure 1. Error bars show pointwise 95% confidence intervals with standard errors clustered by stock and year-month.

around these cutoffs to identify the causal effect of changes in BMI on CAPM $\widehat{\beta}$.[12]

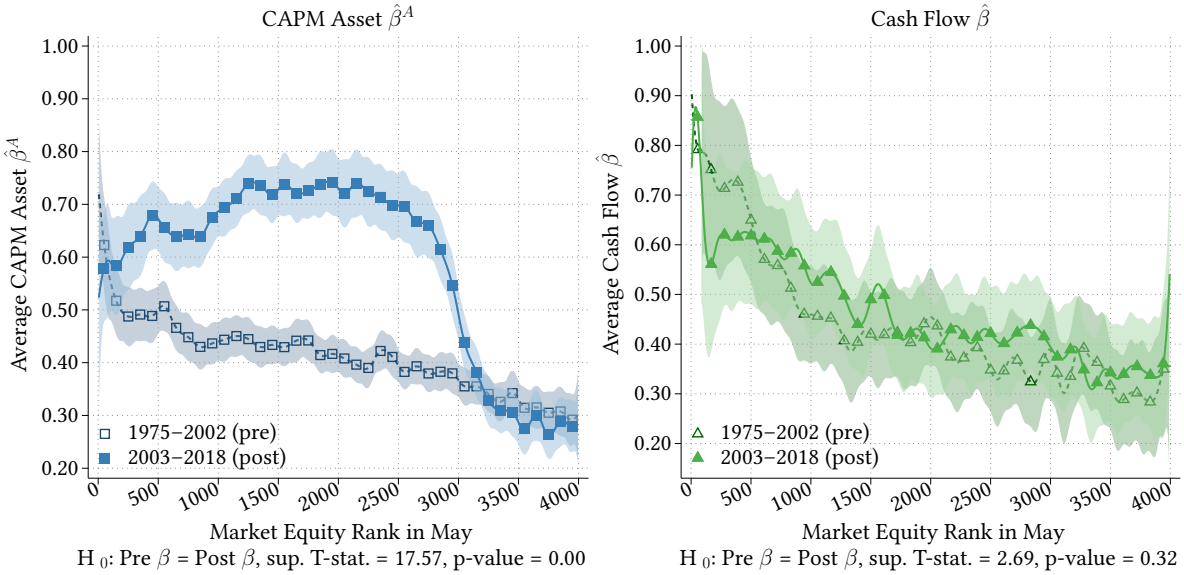
Fact 2: The increases in CAPM $\widehat{\beta}$s are not driven by changes in firms' capital structure. A potential confounder is that changes in firms' capital structure could drive observed changes in CAPM equity $\widehat{\beta}$s rather than benchmarking. Although it seems unlikely that systematic changes in capital structure would explain the average CAPM equity $\widehat{\beta}$ changes within an economically arbitrary rank bin, we verify that changes in CAPM equity $\widehat{\beta}$s are unrelated to changes in leverage.

CAPM asset $\widehat{\beta}$s exhibit similar patterns as CAPM equity $\widehat{\beta}$s. The left panel of Figure 3 shows binned scatter plots of stocks' (unlevered) CAPM asset $\widehat{\beta}$s against equity ranks for the periods pre and post-2003. The average CAPM asset $\widehat{\beta}$ of Russell 2000 stocks increased by 0.29. We find no evidence that changes in firms' leverage are driving the observed changes in CAPM $\widehat{\beta}$s.

Fact 3: CAPM $\widehat{\beta}$s increased while fundamental risk exposure stayed constant. The increases in CAPM $\widehat{\beta}$ are unrelated to changes in firms' fundamental risk exposure as measured by cash flow βs. Figure 3 also allows us to compare cash flow βs estimated from accounting data to

[12]The increase in BMI after the 200th rank is due to the Russell Midcap index that tracks stocks smaller than rank 200 within the Russell 1000. The AUM of funds benchmarked to the Russell Midcap is almost as high as that of all Russell 2000 funds (Pavlova and Sikorskaya, 2023).

Figure 3: CAPM Asset $\widehat{\beta}^A$ and Cash Flow β vs. Market Equity Rank in May



Notes: This figure shows binned scatter plots of stocks' quarterly cash flow β and CAPM asset $\widehat{\beta}$ vs. their market equity rank in May. We follow Krüger et al. (2015) and unlever equity βs as $\widehat{\beta}^A = E/(E+D) \times \hat{\beta}^E$ in which E is the market value of equity and D is the total book value of debt. Each bin shows the equal-weighted average of approximately 100 ranks. Conditional means across market equity ranks are identified using only cross-sectional variation by absorbing year-quarter fixed effects. Outlined bins show estimates from 1975 to 2002, while filled bins show estimates from 2003 to 2018. Shaded areas are 95% confidence bands based on standard errors clustered at the stock and year-quarter level.

CAPM asset βs estimated from stock market returns.[13] Cash flow βs remain stable across market equity ranks in both periods, while unlevered CAPM $\widehat{\beta}$s show a significant increase after 2003. This is particularly pronounced for Russell 2000 stocks, for which CAPM asset $\widehat{\beta}$s rose by an average of 0.29, while cash flow βs stayed constant. Notably, Appendix Figure A3 shows that the difference between conditional means of CAPM asset $\widehat{\beta}$s and cash flow βs across equity ranks was statistically insignificant from 1975 to 2002 but became statistically significant after 2003.

3 A Stylized Model that Highlights the Mechanism

With these facts in hand, we present a stylized model that illustrates how distortions in CAPM $\widehat{\beta}$s through their effect on discount rates affect corporate investment decisions. Every standard corporate finance textbook instructs firm managers to implement investment policies that maxi-

[13] We follow Cohen, Polk, and Vuolteenaho (2009) and estimate cash flow βs as $\text{ROE}_{i,t} = \alpha_i + \beta_i^{CF} \text{ROE}_{Mkt,t} + \varepsilon_{i,t}$. ROE denotes the ratio of clean-surplus earnings ($X_t = BE_t - BE_{t-1} + D_t$) to beginning-of-the-period book equity (BE_{t-1}). D_t are gross dividends computed from the difference between CRSP returns and returns excluding dividends. We extend our analysis to the period from 1975 to 2018 since accounting information is only available quarterly and compute β_i^{CF} separately for each firm in Compustat using an expanding window of observations.

mize firm value. The canonical guidance is to maximize net present value (NPV).[14] Firm managers need two key components to calculate NPV: the expected cash flows and the discount rate to convert future cash flows to present value. Most finance textbooks recommend using the weighted average cost of capital (WACC) as discount rate and to estimate the CAPM β to determine the cost of equity. The following stylized model highlights how the presence of benchmarked funds distorts firms' discount rates and thus affects capital allocation decisions.

Textbook investment policy Firm value $V_{i,t}$ is determined by the net present value of its expected future cash flows $\{CF_{i,t+h}\}_{h=1}^{\infty}$ discounted at the firm-specific discount rate $\{R_{i,t\to t+h}\}_{h=1}^{\infty}$:

$$V_{i,t} = \mathbb{E}_t\left[\sum_{h=1}^{\infty} \frac{CF_{i,t+h}}{R_{i,t\to t+h}}\right]. \tag{2}$$

The discount rate $R_{i,t\to t+h}$ equals the firm's weighted average cost of capital, determined by exposure to aggregate risk of the cash flows generated by the firm's assets, β_i^A, and the yield curve of risk-free rates $\{R_{t+h}^f\}_{h=1}^{\infty}$.

$$R_{i,t\to t+h} = \prod_{j=1}^{h}\left(R_{t+j}^f + \beta_i^A ERP_{t+j}\right) \tag{3}$$

Assuming for simplicity that firm leverage remains constant over time and is sufficiently low to not create default risk, the aggregate risk exposure of the firm's cash flows is proportional to exposure of the firm's equity to the equity risk premium (ERP_t):

$$\beta_i^A = \frac{\beta_i^E}{1 + (1-\tau)\frac{D_i}{E_i}} \tag{4}$$

As such, it can be directly inferred from the empirical CAPM β of the firm's equity $\widehat{\beta}_i^E = \frac{\widehat{\text{Cov}}(r_i, r_m)}{\widehat{\text{Var}}(r_m)}$. A firm manager considering a firm-typical project with cost $C_{i,t}$ and future cash flows $\{y_{i,t+h}\}_{h=1}^{\infty}$ should invest in the project if it has positive NPV and thus increases firm value, that is if

$$\mathbb{E}_t\left[\sum_{h=1}^{\infty} \frac{y_{i,t+h}}{\widehat{R}_{i,t+h}}\right] = \mathbb{E}_t\left[\sum_{h=1}^{\infty} \frac{y_{it+h}}{\prod_{j=1}^{h}\left(R_{t+j}^f + \frac{\widehat{\beta}_i^E}{1+(1-\tau)\frac{D_i}{E_i}} ERP_{t+j}\right)}\right] > C_{i,t}, \tag{5}$$

[14] Graham, Harvey, and Puri (2015) show that CEOs and CFOs prioritize NPV in capital allocation decisions.

in which $\widehat{R}_{i,t}$ results from substituting the empirical counterpart of (4) into (3).

The presence of benchmarked funds distorts asset prices. Kashyap et al. (2021) show that benchmarked funds' inelastic demand for benchmark constituents increases their price and thereby lowers their implied discount rate. However, benchmark membership also induces excess co-movement between constituents that is unrelated to the aggregate risk exposure of the firm's cash flows (Vijh, 1994, Barberis et al., 2005, Boyer, 2011).[15] The excess co-movement of benchmark stocks increases $\widehat{\beta}_i^E$ and thus the discount rate proportional to the equity risk premium.

We illustrate the two opposing effects in reduced form by postulating two discount rate wedges as functions of a stock's benchmarking intensity (BMI). The price pressure from benchmark inclusion reduces the implied discount rate by $\Delta_I(BMI_{i,t})$. At the same time, β^E increases to $\widehat{\beta}_{i,t}^E = \beta_i^E + \Delta_\beta(BMI_{i,t})$. Both $\Delta_I(BMI_{i,t})$ and $\Delta_\beta(BMI_{i,t})$ monotonically increase in $BMI_{i,t}$ and satisfy $\Delta_I(0) = \Delta_\beta(0) = 0$.

The discount rate, adjusted for benchmarking distortions, that maximizes firm value is

$$\widetilde{R}_{i,t\to t+h} = \prod_{j=1}^{h} \left(R_{t+j}^f + \frac{\beta_i^E + \Delta_\beta(BMI_{i,t})}{1 + (1-\tau)\frac{D_i}{E_i}} ERP_{t+j} - \Delta_I(BMI_{i,t}) \right). \tag{6}$$

CAPM investment policy with benchmarking distortions Whether the benchmarking distortions to the discount rate in Eq. (6) cause firm managers to invest more or less is ambiguous. The price effect of benchmark inclusion, $\Delta_I(BMI_{i,t})$ (in blue) incentivizes investment, whereas the increase in CAPM $\widehat{\beta}$, Δ_β (in red), discourages investment. Empirically, positive price effects at inclusion (see e.g., Chang et al., 2015) suggest that $\Delta_I(BMI_{i,t})$ may dominate in the short-run.

What if firm managers do not internalize the inelastic demand of benchmarked funds for index constituent stocks but instead follow textbook guidance and use the cost of capital implied by the CAPM to evaluate investment opportunities as in Eq. (5)? In this case, the presence of benchmarked funds has an unambiguously negative effect on investment for benchmark constituents. Firm managers, with subjective expectations $\mathbb{E}_t^\star[\cdot]$, observe an increase in their stock's CAPM β from β_i^E to $\widehat{\beta}_i^E = \beta_i^E + \Delta_\beta(BMI_{i,t})$ and infer an increase in the firm's cost of capital. Firm managers now invest only in projects that satisfy

$$\mathbb{E}_t^\star \left[\sum_{h=1}^{\infty} \frac{y_{i,t+h}}{\widetilde{R}_{i,t+h}} \right] = \mathbb{E}_t^\star \left[\sum_{h=1}^{\infty} \frac{y_{i,t+h}}{\prod_{j=1}^{h} \left(R_{t+j}^f + \frac{\beta_i^E + \Delta_\beta(BMI_{i,t})}{1+(1-\tau)\frac{D_i}{E_i}} ERP_{t+j} \right)} \right] > C_{i,t}. \tag{7}$$

[15] See also Appendix D of Kashyap et al. (2021) for further details.

All else equal, a firm inside a benchmark index invests less than the same firm outside the index.

Testable hypothesis Our proposed mechanism rests on the behavioral assumption that firm managers do not internalize the total effect of benchmarking-induced discount rate distortions. Instead, managers follow textbook guidance to form a weighted average cost of capital using their firm's empirical CAPM $\widehat{\beta}$s. Benchmarking-induced co-movement distorts the CAPM $\widehat{\beta}$s and leads benchmark constituents to under-invest.

We empirically validate our mechanism by presenting evidence that supports three testable hypotheses directly derived from it. All else equal,

(i) there is a monotonic positive relationship between changes in firm BMI and changes in $\widehat{\beta}^{E}$,
(ii) an increase in firm BMI increases the firm's perceived cost of capital,
(iii) and leads to a decline in investment.

4 Data and Sample

We use four main sources in our empirical analysis: (1) the measure of benchmarking intensity developed by Pavlova and Sikorskaya (2023), (2) estimates of firm CAPM $\widehat{\beta}$s from Welch (2022b), (3) firm-level data from S&P's Compustat North America Fundamentals, and (4) additional firm-level variables from various sources.

Benchmarking Intensity We use monthly estimates of benchmarking intensity from Pavlova and Sikorskaya (2023) which cover 1998 to 2018. Similar to Pavlova and Sikorskaya (2023), we use changes in BMI between the rank day of Russell indices in May and the reconstitution day in June as an instrumental variable for changes in institutional ownership.[16] Changes in BMI satisfy the relevance condition because they predict how benchmarked investors rebalance their portfolios after a Russell index reconstitution. The exclusion restriction requires that index membership is exogenous. The literature on benchmark inclusion effects argues that after controlling for factors that determine benchmark inclusion, most importantly the ranking variable (market value) that Russell uses for index assignment at the end of May, the index membership is exogenous.

Estimates of CAPM βs We use end-of-month estimates of the winsorized and exponentially weighted CAPM $\widehat{\beta}$s from Welch (2022b), calculated using daily data against the CRSP value-weighted index. The β estimator first winsorizes a firm's daily stock return at −2 and +4 times the contemporaneous market return, and then estimates an exponentially weighting least squares

[16] For the average stock, approximately 72.6% of BMI is contributed by funds which are benchmarked to Russell indices while 10.7% come from S&P 400/500 and 8.4% from CRSP indices. The remainders consists of other index providers such as Dow Jones, FTSE, and Wilshire.

regression of the stock's (winsorized) excess return on the market excess return. The half-life of the exponential decay is 90 trading days. Welch (2022b) shows that this estimator outperforms other β estimators like Bloomberg's β or the Vasicek (1973) β in predicting future market βs out of sample. We focus on common equities that trade on either the NYSE, AMEX, or NASDAQ, and exclude ADRs, REITs, and ETFs. Appendix Table A1 reports descriptive statistics for the matched BMI-CAPM data covering 1998 to 2019.

Firm-level data We use annual data for publicly listed companies incorporated and located in the U.S. from Compustat for the sample period from 1998 to 2018. We exclude financial firms (SIC codes 6000-6999) and firms in regulated industries (4900-4999), as well as firms with less than \$50m in total assets or less than \$10m sales (in 2017 dollars). Firms must have at least five years of consecutive data. We winsorize the data at the 2.5% and 97.5% level.

Additional firm data We obtain stock returns and market data from the Center for Research in Security Prices (CRSP). We source data on firm managers' perceived cost of capital from Gormsen and Huber (2024), who hand-collect these from earnings calls. We use measures of firm-level intangible capital from Peters and Taylor (2017). We collect Executive compensation peer group information from Institutional Shareholder Services (ISS). We use firm-level risk measures from Hassan, Hollander, Van Lent, and Tahoun (2019) and data on financial frictions from Hoberg and Maksimovic (2015) and Linn and Weagley (2021). We obtain Governance scores from Sustainalytics, Refinitiv, and S&P and estimates of stocks' implied cost of capital (ICC) from Eskildsen et al. (2024).

5 Changes in Stocks' CAPM βs Caused by Benchmarking

We estimate the causal effect of changes in a firm's benchmarking intensity on its CAPM equity $\widehat{\beta}$ using a difference-in-differences design. We compare the evolution of CAPM $\widehat{\beta}$s of (treated) stocks that experience BMI changes around Russell index reconstitution dates to (control) stocks that do not. The results show that changes in a stock's BMI around index reconstitutions causally change its CAPM $\widehat{\beta}$. This effect is symmetric and monotonically increasing in treatment intensity. A stock whose BMI increases (decreases) by at least 5 p.p. experiences an increase (decrease) in CAPM $\widehat{\beta}$ by 0.18 (-0.23) relative to the control group. The change in CAPM $\widehat{\beta}$ persist for at least 7 years and only partly reverses over time.

Managers' perceived cost of capital responds to benchmarking-induced increases in CAPM $\widehat{\beta}$s. Using data from Gormsen and Huber (2024), we find that increases in BMI predict increases in managers' perceived cost of capital. We use changes in BMI as an IV to identify the causal

effect of changes in CAPM $\widehat{\beta}$s on a firm's perceived cost of capital. The IV estimates imply that a 0.20 change in CAPM $\widehat{\beta}$ increases managers' perceived cost of capital by approximately 74 bps.

The effects of increases in benchmarking on CAPM $\widehat{\beta}$s are large and economically significant. An increase of 0.18 in CAPM β translates to a 108 bps rise in the cost of equity capital, assuming a 6% equity risk premium. The bias can have substantial implications for NPV calculations. For instance, a firm investing in a project that generates $100 in perpetuity would value it at $1,000 with a 10% discount rate. However, increasing the discount rate to 11.1% reduces the project's value to $900. Overestimating the WACC by just 1.08% leads to a nearly 10% undervaluation. We analyze the effects of these distortions on firm and industry investment in Section 6.

5.1 Difference-in-differences Strategy

We analyze the effect of an increase in BMI on a firm's CAPM $\widehat{\beta}$ by estimating a series of difference-in-differences specifications of the form:

$$\text{CAPM } \widehat{\beta}_{i,t} = \delta \text{ Treated}_i \times \text{Post}_{t>\text{May}} + \theta_i + \theta_{t,s} + \zeta X_{i,t} + \varepsilon_{i,t} \tag{8}$$

in which Treated_i is an indicator variable for whether firm i's BMI changed by more than ± 5 p.p.. The coefficient of interest, δ, summarizes the treatment effect on a firm's CAPM $\widehat{\beta}$. We restrict our sample to all stocks stocks within 250 ranks around the Russell index cutoffs to ensure we capture only changes in BMI due to Russell index reconstitution.[17]

The inclusion of firm fixed effects, θ_i, removes time-invariant heterogeneity across firms and accounts for possible ex-ante differences between treated and control firms. Size-by-month or liquidity-by-month fixed effects, $\theta_{s,t}$, restrict the identifying variation to comparisons within the same size or liquidity decile each period, controlling for time-varying unobserved heterogeneity correlated with a firm's CAPM $\widehat{\beta}$. We exclude stocks with market values below the fifth percentile of the NYSE to ensure that micro-cap stocks do not drive the results.[18] We additionally include stock-level controls, $X_{i,t}$, to improve precision. These are the stock's monthly excess return, log of trading volume, as well as the cumulative excess return over the past 12 and 24 months.

The Russell indices reconstitute each year, and identification in (8) thus exhibits staggered treatment timing with potentially misleading estimates if the treatment effect is heterogeneous between cohorts or over time (De Chaisemartin and d'Haultfoeuille, 2023). We follow the suggestion of Baker, Larcker, and Wang (2022) to stack the yearly cohorts and include cohort by firm

[17] Our findings are robust to changing the bandwidth to 100, 300, 500 ranks or not imposing restrictions.
[18] We obtain similar results when using the 10th or 20th percentile of the NYSE as a cutoff (see Appendix Table A2).

and cohort by time fixed effects but suppress cohort subscripts for brevity.[19]

Identifying assumptions and threats to identification. Our identification strategy relies on the following assumption: conditional on the set of fixed effects and control variables, firms that experience changes in BMI are not differentially exposed to unobservable shocks that correlate with BMI. The identification assumption does not require random assignment of benchmark inclusion, nor does it require that firms have similar characteristics in levels. Rather, we rely on the parallel trends assumption that CAPM $\widehat{\beta}$s for treated and control firms would have trended similarly absent an increase in BMI. We provide supporting evidence of parallel pre-treatment trends in Figures 4a and 4b and additionally perform a placebo test in Figure A6. As expected, this placebo test finds no treatment effect.

Results Figures 4a and 4b show the event study coefficients of our difference-in-differences estimation for treatments defined as an increase or decrease in BMI of 5 p.p., respectively. We normalize the dynamic treatment effect to zero in May and estimate dynamic treatment effects for the period from 9 months before to 12 months after benchmark inclusion.

Several facts are worth noting. First, the CAPM $\widehat{\beta}$s of treated and control firms evolve in close parallel before index reconstitution, supporting the identification assumption and causal interpretation of treatment effects. The p-values for the Wald-style pre-trend test suggested by Freyaldenhoven, Hansen, Pérez, and Shapiro (2021) are 0.98 and 0.95, respectively. Second, after index reconstitution, CAPM $\widehat{\beta}$s begin to diverge for both increases and decreases in BMI. The smooth event-time trend in Figure 4 is largely mechanical as older information receives exponentially smaller weights.[20] The treatment effect of a BMI increase on a firm's CAPM $\widehat{\beta}$ is likely immediate, but our measurement captures it only once older information is sufficiently down-weighted. Third, treatment effects for BMI increases and decreases of at least 5 p.p. have similar magnitudes but opposite signs, suggesting that treatment effects are linear in BMI.

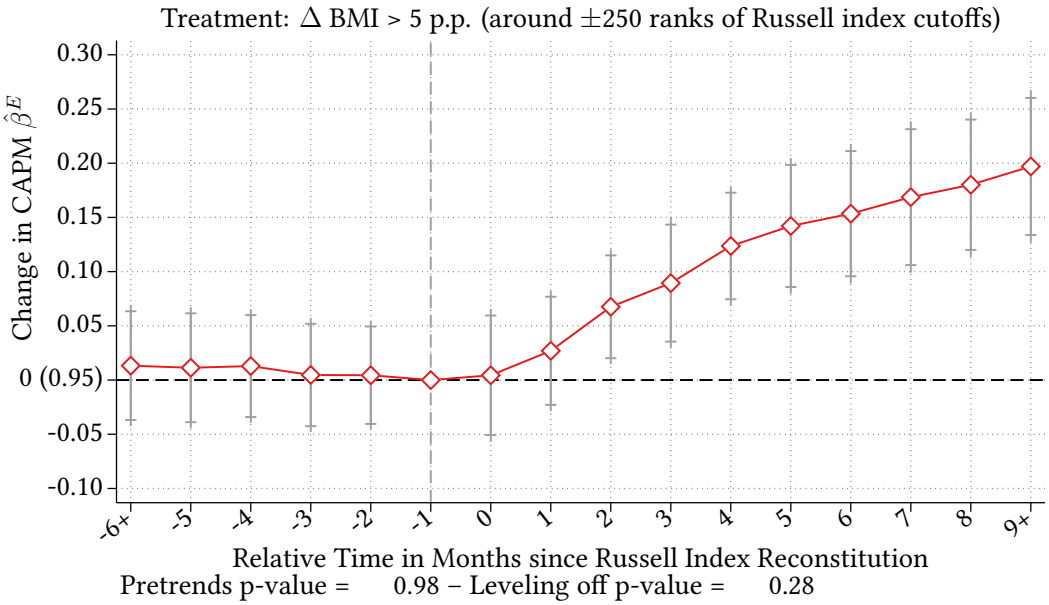
Table 1 reports the average estimated post-shock coefficient.[21] The coefficient estimate for the interaction $\text{Treated}_i \times \text{Post}_{t>\text{May}}$ is positive for increases in BMI and negative for decreases in BMI, with treatment effects always statistically significant at the 0.1% level. Columns (1) and (5) present results with firm and year-month fixed effects, while subsequent columns add more granular fixed effects by year-month, market equity rank, volume, or shares outstanding. These fixed effects control for potential size or liquidity differences between treated and control firms.

[19] See, e.g., Gormley and Matsa (2011), Cengiz, Dube, Lindner, and Zipperer (2019) and Deshpande and Li (2019) for similar implementations of stacked difference-in-differences designs.

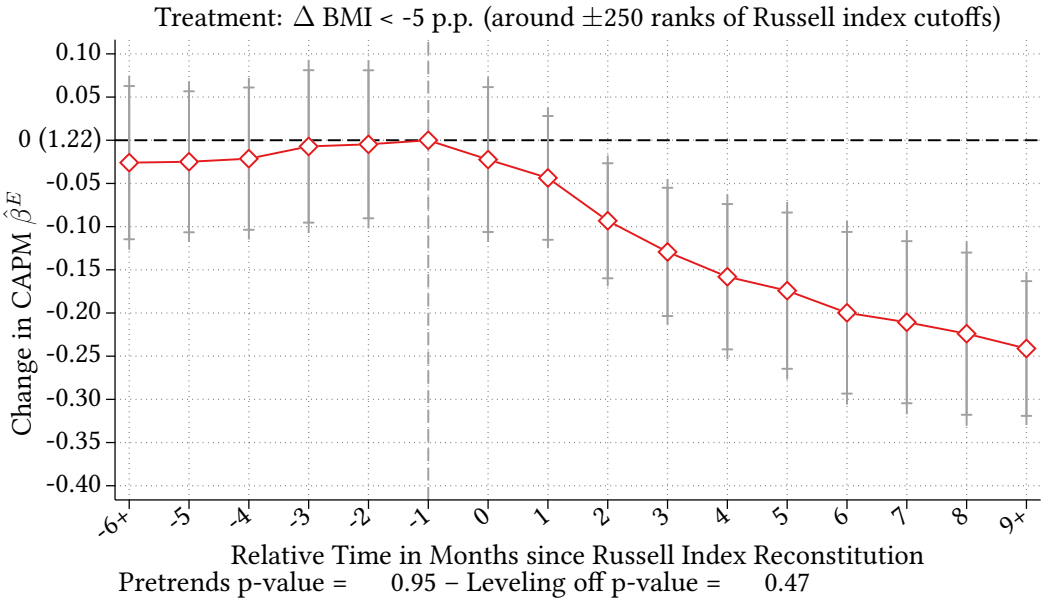
[20] The β estimator uses an expanding-window with exponential weights of 4.5 months half-life.

[21] We exclude the coefficients for $0 \leq t \leq 6$ since the CAPM $\widehat{\beta}$s by construction have a dynamic component.

Figure 4: Difference-in-differences Event Study of Changes in BMI on Changesn in CAPM $\widehat{\beta}^E$



(a) Treated stocks with an increase in Δ BMI > 5 p.p. relative to control group.



(b) Treated stocks with a decrease in Δ BMI < -5 p.p. relative to control group.

Notes: This figure shows difference-in-differences event study coefficients of Eq. (8). Change in CAPM $\widehat{\beta}^E$ of treated stocks with an increase or decrease in BMI of at least 5 p.p. relative to a control group, in which $|\Delta \text{ BMI}| < 1$ p.p.. Pointwise confidence intervals (99%) and sup-t confidence bands based on double-clustered standard errors. Values in parentheses on the Y-axis show the average CAPM $\widehat{\beta}^E$ before treatment.

Table 1: Causal Effects of Changes in Benchmarking Intensity on CAPM $\widehat{\beta}$s

Treatment Group:	Δ BMI > 5 p.p.				Δ BMI < -5 p.p.			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Treated × Post	0.183***	0.182***	0.185***	0.186***	-0.221***	-0.225***	-0.218***	-0.218***
	(0.021)	(0.021)	(0.021)	(0.019)	(0.026)	(0.030)	(0.025)	(0.026)
Momentum and Liquidity Controls	✓	✓	✓	✓	✓	✓	✓	✓
Fixed Effects								
Firm × Cohort	✓	✓	✓	✓	✓	✓	✓	✓
Time × Cohort	✓				✓			
Rank Decile × Time × Cohort		✓				✓		
Volume Decile × Time × Cohort			✓				✓	
Shrs. Out. Decile × Time × Cohort				✓				✓
Observations	90,655	90,640	90,568	90,593	91,928	91,898	91,832	91,833

Notes: This table reports $\hat{\delta}$ for specifications of the form: CAPM $\widehat{\beta}_{i,t} = \delta$ Treated$_i$ × Post$_{t>\text{May}} + \theta_i + \theta_{t,s} + \varepsilon_{i,t}$. Treated×Post is average of post-treatment coefficients after 7 months (to account for the expanding-window estimation of CAPM $\widehat{\beta}$). Control group are firms with |ΔBMI| <1 p.p.. Estimation sample is restricted to stocks within 250 ranks around Russell index cutoffs. Standard errors in parentheses are double-clustered at stock and year-month level. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001.

The estimated effects of BMI changes remain stable across different fixed effects. Column (2) shows that a BMI increase of at least 5 p.p. raises a firm's CAPM $\widehat{\beta}$ by 0.18, while Column (6) shows that a decrease in BMI by at least 5 p.p. lowers a firm's CAPM $\widehat{\beta}$ by -0.23.

The estimates of the average treatment effect in Table 1 mask substantial heterogeneity. Appendix Figure A5 plots the event-time coefficients for three different treatment intensities. Treatment levels are defined in increasing order of treatment intensity as changes in BMI by 5 p.p. to 10 p.p., 10 p.p. to 20 p.p., and greater than 20 p.p.. The estimated treatment effect is 0.15 for the lowest treatment intensity, 0.24 for the medium treatment intensity, and 0.35 for the highest treatment intensity after 12 months. The treatment effects are statistically significant at the 0.1% level for all treatment intensities. For firms in the highest treatment intensity group, the treatment effect is 0.35, equivalent to a 210 bps increase in the cost of equity capital.

Alternative CAPM β Estimators To verify that our results are not driven by the choice of β estimator, we estimate a series of OLS regressions for a set of alternative estimators. Appendix Table A3 shows OLS regressions results using different estimators for the unknown CAPM β. We compare the usual OLS estimator with alternative estimators proposed by Blume (1975), Dimson (1979), and Welch (2022b). We also report estimates of the $\widehat{\beta}$ with the ten largest stocks by market capitalization, and correlation with the market and idiosyncratic volatility. We find that *all* CAPM $\widehat{\beta}$ estimators show a statistically significant (p<0.001) association with changes in BMI.[22] Changes in CAPM $\widehat{\beta}$ occur due to changes in market correlation, not because of increases in idiosyncratic volatility. This aligns with Antón and Polk (2014), who show that mutual funds'

[22]Online Appendix Table B1 shows that industry CAPM $\widehat{\beta}$s estimates react similar.

common ownership increases return correlation of equities.

5.2 Persistence of Effects and Changes in Stock's Implied Cost of Capital

We assume that managers use the CAPM to set discount rates and do not account for the distorting effects of benchmarking on stock prices. Alternatively, managers might infer the discount rate from stock prices and expected cash flows, as in Kashyap et al. (2021). We test whether BMI changes at index reconstitution influence the implied cost of capital (ICC) that managers could infer from stock prices. The ICC, based on current stock prices, captures any price distortion from benchmarking. Following Eskildsen et al. (2024),[23] we calculate the ICC by averaging four popular models from the accounting literature: the residual income models from Gebhardt et al. (2001) and Claus and Thomas (2001), and the dividend discount models from Easton (2004) and Ohlson and Juettner-Nauroth (2005). The ICC only captures the implied cost of equity, we thus compare it to the cost of equity implied by the CAPM.

We examine whether benchmarking distortions persist over long horizons. Our difference-in-differences analysis shows that BMI distortions in CAPM $\widehat{\beta}$s persist for at least 12 months. If these distortions in the cost of equity fade after a year, long-term effects on investment are unlikely. We extend our analysis to test for effects on the ICC and CAPM $\widehat{\beta}$s over a seven-year horizon using the following specifications:

$$\text{Avg. ICC}_{i,t+h} = \theta_0^h + \theta_1^h \Delta\text{BMI}_{i,t} + \xi^h X_{i,t} + \varepsilon_{i,t+h}, \quad h = 0, \ldots, 7, \tag{9}$$

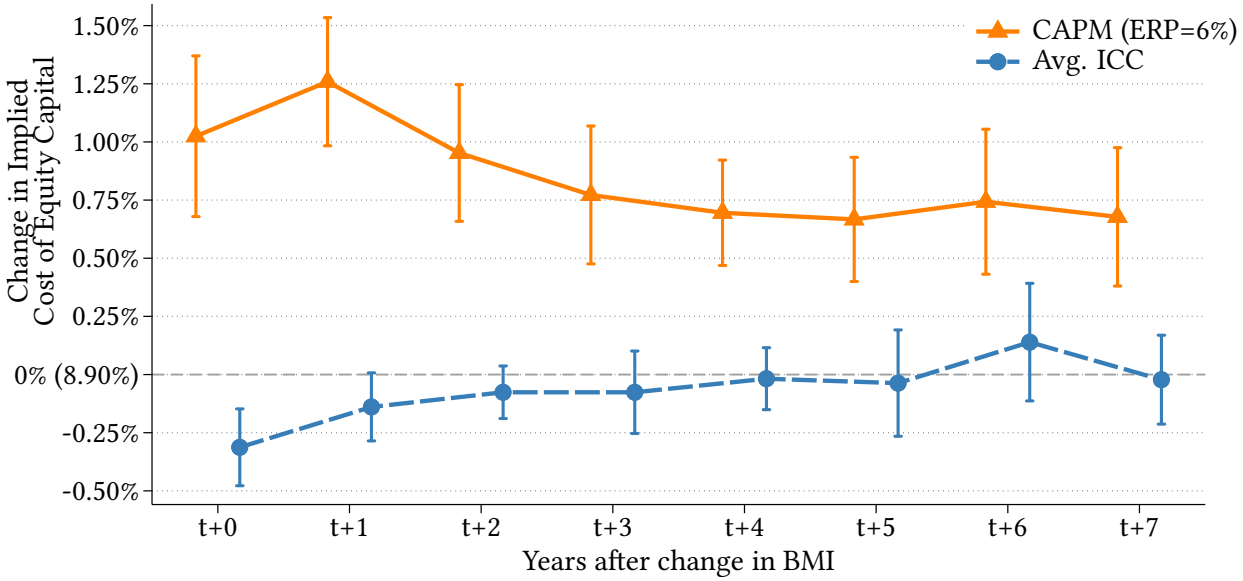
$$\text{CAPM } \widehat{\beta}_{i,t+h} = \gamma_0^h + \gamma_1^h \Delta\text{BMI}_{i,t} + \zeta^h X_{i,t} + \nu_{i,t+h}, \quad h = 0, \ldots, 7. \tag{10}$$

for firm i in year t+h. The coefficients of interest, γ_1^h and θ_1^h, summarize the long-term effects of an BMI increase on a firm's CAPM $\widehat{\beta}$ or ICC after h years, respectively. The vector $X_{i,t}$ contains year-by-industry fixed effects and the lagged level of BMI. We scale the estimates to a 10 p.p. increase in BMI for ease of interpretation and adjust the CAPM estimates to match the units of the ICC estimates by multiplying them by a 6% ERP. We focus on stocks in the 300 ranks around the Russell index cutoffs.

Results Figure 5 shows estimates for γ_1^h and θ_1^h of Eq. (10) and (10), respectively. The distortionary effects of BMI increases on CAPM $\widehat{\beta}$s persist for at least 7 years. At each forecast horizon, we find statistically significant (p<0.001) positive effects of BMI increases on CAPM $\widehat{\beta}$s. These effects gradually diminish over time but remain economically significant. One year after the ini-

[23] Using these measures, Eskildsen et al. (2024) estimate an annual green equity premium of -25 bps. Similarly, Kontz (2023) estimates a green convenience yield of 28 bps in auto asset-backed securities.

Figure 5: Persistence of BMI Shock on the Cost of Equity Capital over Long Horizons



Notes: This figure shows the persistence of BMI shocks on estimates of firms' cost of equity for ICC and CAPM using regression of the form: CAPM $\widehat{\beta}_{i,t+h} = \gamma_0^h + \gamma_1^h \Delta\text{BMI}_{i,t} + \varepsilon_{i,t+h}$. Estimation sample is restricted to stocks within 250 ranks around Russell index cutoffs. Estimates are scaled to a 10 p.p. BMI increase. 99% confidence intervals based on standard errors clustered at stock and year. Value in parentheses on the Y-axis is the median ICC over the sample.

tial BMI increase, the cost of equity capital is 125 bps higher. Four years later, it remains 71 bps higher, and even seven years later, it is still 67 bps higher. This prolonged impact suggests that benchmarking has a long-term effect on firms' perceived cost of capital, potentially leading to sustained changes in investment behavior.

In contrast, changes in BMI have only short-lived effects on the implied cost of capital. The estimates for the year of benchmark inclusion show that the ICC decreases by approximately 36 bps (p<0.001). We perform a back-of-the-envelope calculation to estimate the implied stock return at benchmark inclusion using Gordon's growth model.[24] The 36 bps decrease in the ICC at benchmark inclusion implies a 5.24% increase in the stock price, close to the 5% documented by Chang et al. (2015). Our estimate is larger but not statistically different from the 2.8% increase in stock price for a 10 p.p. increase in BMI estimated by Pavlova and Sikorskaya (2023). However, the effect on the ICC fades to 14 bps (p<0.05) after one year and becomes statistically insignificant afterwards. The impact of BMI increases on the ICC is small and diminishes quickly. This makes it unlikely that BMI changes affect investment through the ICC.

[24] We assume the expected dividend, D_1, and expected dividend growth rate, g, remain constant when BMI changes, but r changes by $\hat{\theta}_1^0 \times \Delta\text{BMI}$. The implied stock return is given by: $\frac{P^{Post}}{P^{Pre}} - 1 = \frac{-\hat{\theta}_1^0 \times \Delta\text{ BMI}}{r + \hat{\theta}_1^0 \times \Delta\text{ BMI} - g} = \frac{0.36\%}{8.9\% - 0.36\% - 1.6\%} \approx 5.24\%$. We set r to the pre-benchmark inclusion average of the ICC (≈8.9%), g to the long-run average of real dividend growth (≈1.6%), and ΔBMI to 10 p.p (as in Figure 5).

5.3 Changes in CAPM β and Manager's Perceived Cost of Capital

We document that changes in the CAPM $\widehat{\beta}$ affect the perceived cost of capital of firm managers, using firm's self-reported data collected from earnings calls by Gormsen and Huber (2023). We estimate the pass-through of changes in a firm's CAPM $\widehat{\beta}$ on the perceived cost of capital using a series of instrumental variable (IV) regressions of the following form:

$$\Delta\text{CAPM } \beta_{i,t} = \delta_i + \delta_{j,t} + \theta \Delta\text{BMI}_{i,t} + \epsilon_{i,t} \tag{11}$$

$$\Delta\text{Perceived Cost of Capital}_{i,t} = \alpha_i + \alpha_{j,t} + \sum_{h=0}^{4} \gamma_h \Delta\widehat{\text{CAPM}}\,\beta_{i,t-h} + \varepsilon_{i,t} \tag{12}$$

in which the coefficient of interest, $\sum_{h=0}^{4} \gamma_h$, provides a causal estimate of the pass-through of changes in a firm's CAPM $\widehat{\beta}$ to its perceived cost of capital. Including firm fixed effects α_i ensures we remove time-invariant heterogeneity across firms, and, in particular, accounts for possible ex-ante differences in characteristics between treated and control firms. Size-by-year or industry-by-year fixed effects, $\alpha_{j,t}$, further restrict the identifying variation to comparing firms within the same size decile, liquidity decile, or industry each period.

We use a distributed lag model in Eq. (12) for both economic and econometric reasons. Managers typically estimate CAPM $\widehat{\beta}$ using OLS regressions using two to five years windows. Rolling-window estimators gradually incorporate changes in CAPM $\widehat{\beta}$ into managers' perceived cost of capital as new data becomes available. Gormsen and Huber (2023) show that this perceived cost of capital affects managers' required returns on new investments and investment decisions, with a transmission lag of several years.[25]

A firm's perceived cost of capital and its stock's CAPM β are jointly determined by its exposure to aggregate risk in equilibrium. To address endogeneity and measurement error, we use an instrumental variable approach, with BMI change as a natural instrument. BMI changes induce CAPM $\widehat{\beta}$ changes orthogonal to aggregate risk as shown by the difference-in-differences results. The exclusion restriction requires that BMI increases affect a firm's perceived cost of capital only through changes in its CAPM β. This could be violated if BMI impacts perceived cost through other channels, like reducing credit spreads due to reputational effects. However, in Appendix B, we find no evidence that BMI changes correlate with risk, financial constraints, or governance changes, supporting the validity of the exclusion restriction.

Table 2 reports coefficient estimates of Eq. (12) for reduced form (RF), OLS, and IV specifica-

[25] Additionally, the data supports a distributed lag model over a contemporaneous effect model, with Bayesian information criterion (BIC) values favoring fourth-order lags (see Appendix Table A4).

Table 2: Effect of ΔCAPM $\widehat{\beta}$ on Managers' Perceived Cost of Capital (Gormsen and Huber, 2024)

	RF			OLS			IV		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Dependent variable: Δ Perceived Cost of Capital								
Δ BMI (in p.p.)	0.0164***	0.0168***	0.0167***						
	(0.004)	(0.004)	(0.004)						
Δ CAPM β^A				1.202***	1.183***	1.135***	3.153***	3.694***	3.148***
				(0.057)	(0.059)	(0.075)	(0.825)	(0.920)	(0.769)
Fixed Effects									
Firm	✓	✓	✓	✓	✓	✓	✓	✓	✓
Time	✓			✓			✓		
Size Quartile × Time		✓			✓			✓	
Industry × Time			✓			✓			✓
Observations	19,501	19,501	18,940	19,501	19,501	18,940	19,501	19,501	18,940

Notes: This table reports $\sum_{h=0}^{4} \hat{\gamma}_h$ for specifications of the form: ΔPerc. Cost of Capital$_{i,t} = \alpha_i + \alpha_{j,t} + \sum_{h=0}^{4} \gamma_h \Delta$CAPM $\beta_{i,t-h} + \varepsilon_{i,t}$ for reduced form, OLS, and IV regression in which the instrument is ΔBMI for stock i in year t. IV estimated via LIML. Yearly sample from 2000 to 2018. Standard errors in parentheses are clustered at firm-level. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001.

tions. Columns (1) to (3) report RF, columns (4) to (6) OLS, and columns (7) to (9) IV estimates. The coefficients are positive, stable across specifications, and statistically significant (p<0.001). Columns (1) to (3) of Table 2 show that increases in BMI predict increases in managers' perceived cost of capital. OLS estimates show a downward bias relative to IV estimates, likely due to classical measurement error. IV estimates are approximately three times larger than OLS estimates. Two mechanisms introduce measurement error. First, we do not observe how firms estimate their CAPM $\widehat{\beta}$ when calculating their perceived cost of capital. We use the CAPM $\widehat{\beta}$ of Welch (2022b) due to their superior performance in predicting future CAPM $\widehat{\beta}$s out of sample. In practice, firms often simply estimate CAPM $\widehat{\beta}$s using equally-weighted observations on a rolling window of two to five years (Berk and DeMarzo, 2023). Second, any empirical CAPM $\widehat{\beta}$ is an *estimate* of the true CAPM β and thus subject to measurement error.

6 Effects of CAPM β Distortions on Capital Accumulation

Our second set of results estimates how firms react to changes in their CAPM β induced by changes in BMI. For a firm manager who follows textbook guidance to set investment policies using the CAPM, an increase in CAPM $\widehat{\beta}$ raises the user cost of capital and should lead to a decline in investment. We therefore test whether changes in CAPM $\widehat{\beta}$ affect firm outcomes like capital expenditure, physical and intangible capital stocks, cash holdings, payouts, and employment.

The firm-level results show that firms react to BMI-induced changes in their CAPM $\widehat{\beta}$ by reducing investment. Specifically, capital expenditure declines by 10.0% over six years, and physi-

cal and intangible capital stocks are 7.1% and 8.4% lower, respectively. Firms initially accumulate cash and then increase payouts to shareholders. The findings are robust to the inclusion of other known predictors of investment like firm size, cash flow, and Tobin's Q.

We find supporting evidence at the industry-level: industries with higher CAPM βs due to higher benchmarking intensity have lower capital accumulation from 2000 to 2016. The results are robust to the inclusion of industry pre-trends and sectoral fixed effects.

The capital accumulation results at firm- and industry-level are consistent with the predictions of our augmented neoclassical investment model, in which firms adjust investment in response to changes in their *perceived* cost of capital implied by the CAPM.

6.1 Effects of Increased Benchmarking at the Firm-level

We use an instrumental variable (IV) local projections (LP) strategy to forecast the effects of a change in CAPM $\widehat{\beta}$ on capital allocation over horizons of up to 6 years.[26] We instrument changes in CAPM $\widehat{\beta}$ with plausibly exogenous changes in benchmarking intensity caused by Russell index reconstitutions to identify the causal effects of benchmarking distortions on investment.

To analyze the effect of BMI-induced changes in a firm's CAPM $\widehat{\beta}$ on real outcomes, we estimate a series of local projection instrumental variable regressions of the following form:

$$\Delta\text{CAPM } \beta_{i,t} = \delta_i + \delta_{j,t} + \theta \Delta\text{BMI}_{i,t} + \zeta X_{i,t} + \epsilon_{i,t} \tag{13}$$

$$\log\left(Y_{i,t+h}\right) - \log\left(Y_{i,t-1}\right) = \alpha_i^h + \alpha_{j,t}^h + \gamma^h \Delta\widehat{\text{CAPM}}\, \beta_{i,t} + \xi^h X_{i,t} + \varepsilon_{i,t+h} \tag{14}$$

for firm i in industry j in calendar year $t+h$. The coefficients of interests, γ^h, provide cumulative local average treatment effects in % after $h = 0, 1, \ldots, 6$ years.

We remove time-invariant heterogeneity across firms by including firm-fixed effects α_i and δ_i in both first and second stage. We additionally include (3-digit SIC) industry-by-year-by-total-asset quintile fixed effects $\alpha_{j,t}$ and $\delta_{j,t}$ to control for time-varying unobserved heterogeneity across industries, such as differences in industry-level business cycles, which may be correlated with firm outcomes. The use of industry-by-year-by-total-asset fixed effects forces the parameters of interest, γ^h, to be identified solely from comparing similar sized firms within the same industry. The vector $X_{i,t}$ includes a set of time-varying firm-level control variables, such as log of market equity (size) at the end of May and cumulative 1-year excess returns (momentum). We additionally include up to three lags of the outcome and shock variables.

[26] Similarly, researchers often use LP-IVs to study the effects of monetary policy on investment or asset prices (e.g., in Jordà, Schularick, and Taylor, 2020, Kroen, Liu, Mian, and Sufi, 2021, and Bauer and Swanson, 2023).

We cluster standard errors at the firm-level, which allows for a completely unrestricted specification of the residual covariance matrix in the time-series dimension. This effectively addresses the issue of serial correlation in residuals arising in a local projection framework.

Identifying Assumptions and Threats to Identification The instrumental variable exclusion restriction in a local projection setting differs slightly from the usual one due to the dynamic structure of the problem. Identification requires a contemporaneous and a lead-lag exclusion restriction. The instrument must be uncorrelated with past and future shocks, at least after including control variables. The exclusion restriction requires that assignment of index membership is exogenous and that changes in BMI only affect firm outcomes through changes in CAPM $\widehat{\beta}$.

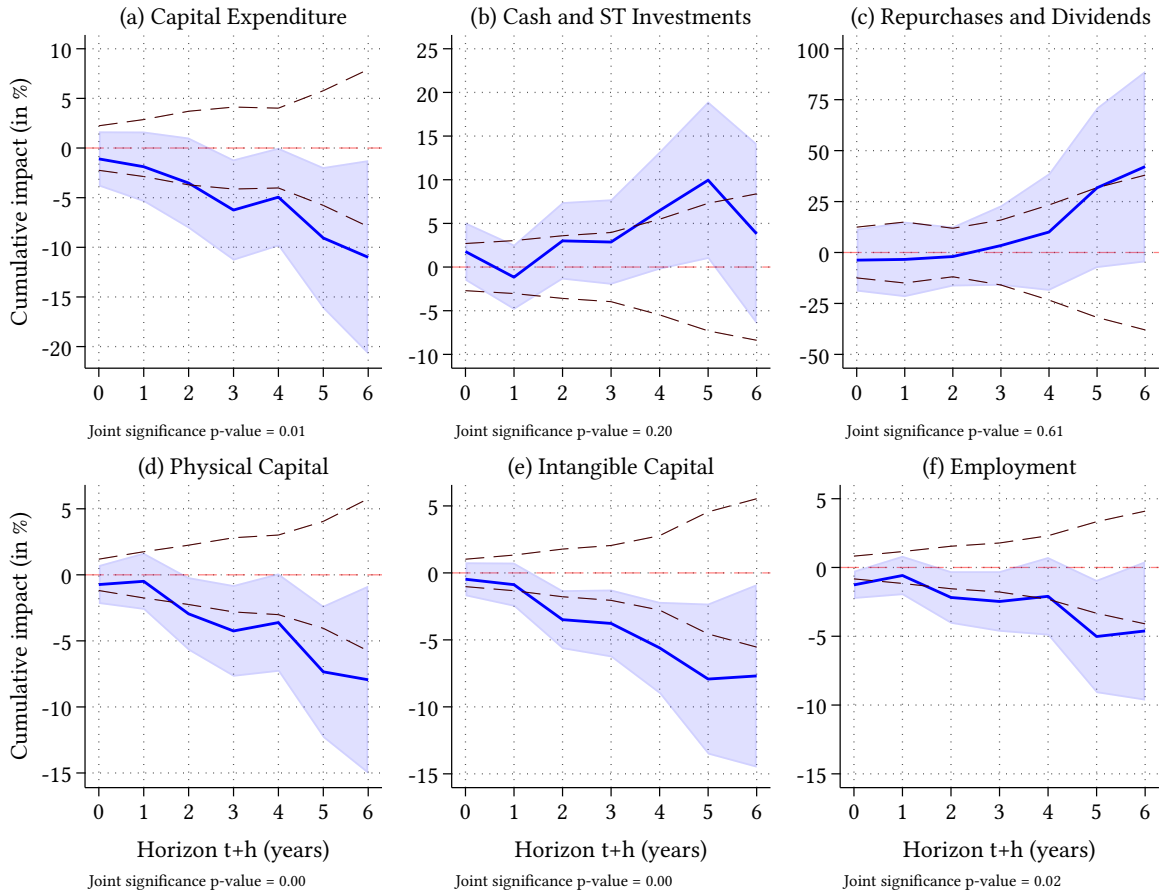
However, concerns may arise that other factors, such as risk exposure, access to debt markets, or governance, could change alongside CAPM $\widehat{\beta}$s when BMI changes, potentially violating the exclusion restriction. In Appendix B, we test whether changes in BMI correlate with changes in firm risk, financial frictions, or governance, but find no evidence that they do. Importantly, Column (1) of Appendix Table A9 shows that changes in BMI do not correlate with firm statements about delaying investments. Failure of the exclusion restriction would introduce bias in the estimated treatment effects. The size and sign of the bias depend on the size and sign of the failure and the strength of the instrument.[27] The literature on benchmark inclusion effects generally finds positive but modest direct effects of inclusion on real outcomes (e.g., Kacperczyk, Sundaresan, and Wang, 2021). We thus, if anything, might expect an upward bias in our estimates.

To ensure that our estimates are well-identified, we follow three steps. First, we include up to three lags of outcomes and shock in our regressions. Second, we saturate our LP-IV estimator with high-dimensional fixed effects to remove as much time-varying unobserved heterogeneity as possible. Third, we verify that including a set of known predictors of capital accumulation (e.g., Tobin's Q or cash flow) does not change our results in a robustness test.

Results Figure 6 shows several key results. First, the impulse responses across all outcome variables have the expected signs: capital expenditure and physical and intangible capital stocks decrease in response to an equity cost shock, while cash holdings increase. Eventually, firms increase dividends and stock repurchases. Employment also decreases, suggesting that firms

[27] Consider the example given by Jordà et al. (2020): let y be the outcome, Δr the intervention, and z the instrument. The IV setup consists of the first and second stage given by $\Delta r = zb + \eta$ and $y = \widehat{\Delta r}\beta + z\phi + \nu$ in which $\mathbb{E}[\Delta r \nu] \neq 0$ but $\mathbb{E}[z\nu] \neq 0$. The exclusion restriction assumes that $\phi = 0$. If $\phi \neq 0$, we have $\widehat{\beta}_{IV} \xrightarrow{p} \beta + \phi/b$. The bias induced by failure of the exclusion restriction depends on both the size of the failure, ϕ, and the strength of the instrument, b. Weaker instruments tend to worsen the bias (see also Conley, Hansen, and Rossi, 2012).

Figure 6: LP-IV: Impulse Response of Outcome Variables to CAPM $\widehat{\beta}$ shock



Notes: This figure shows LP-IV coefficient estimates for $100\times$ cumulative log-changes of outcome variables. Intervention is a $\approx$0.16 shock to CAPM β^E estimated with change in BMI as an instrumental variable. Dashed red lines represent 90% significance bands for the null of zero treatment effect, computed by inverting the F-statistic of joint significance around zero using Scheffé's method (see Jordà, 2023).

reduce labor input in response to an increase in their perceived cost of capital.[28]

Second, firm respond gradually to an increase in CAPM $\widehat{\beta}$, with effects starting from zero in the treatment year and growing over time. The cumulative impact becomes statistically and economically significant after about three years, aligning with industry practices of using a two to five year rolling window to estimate CAPM $\widehat{\beta}$s. This gradual adjustment reflects how managers' estimated cost of capital incorporates older data points. Third, benchmarking induced asset price distortions cause persistent effects on capital expenditure and capital stocks that remain signifi-

[28] This could either be due to labor and capital being complements in firm's production function or due to labor market frictions that prevent employment from freely adjusting in response to shocks. Borovička and Borovičková (2018) argue that fluctuations in discount rates and labor market frictions markets play an important role for employment.

cant for at least six years after the shock.

Benchmarking-induced increases in CAPM $\widehat{\beta}$s lead to large and persistent declines in investment. We scale the shock to the average treatment effect for a BMI increase of at least 5 p.p., which corresponds to a change in the CAPM $\widehat{\beta}$ of 20%. We find that firms reduce their capital expenditure by approximately 10.0% over six years in response to a shock to their CAPM $\widehat{\beta}$ of 20%. The resulting decrease in physical capital stocks is 7.1% and in intangible capital stocks is 8.4% after six years. Point estimates after six years imply a user cost of capital elasticity of physical and intangible capital larger but close to the theoretical elasticity of 1 implied by Cobb-Douglas production.

Robustness Checks We perform several robustness checks on the main results. First, we add known predictors of investment, such as cash flow, Tobin's Q, and the debt-to-equity ratio, to the LP-IV regressions. Second, we incorporate different levels of fixed effects, replacing firm size by industry by year fixed effects with sales by industry by year fixed effects. Appendix Figure A7 shows coefficient estimates of these robustness checks alongside the original estimate. The figure shows that adding firm-level controls does not change the estimates, supporting our identification strategy. Altering the level of fixed effects only marginally affects point estimates, with all changes well within one standard error of the original estimates.

Investment Rates Appendix Figure A8 shows the impact of a 20% increase in CAPM $\widehat{\beta}$ on corporate investment rates. The investment rate[29] declines over time in response to the shock. Starting from a near-zero, gradually becoming more negative, reaching the lowest point after four to five years. The shock to the CAPM $\widehat{\beta}$ leads to a significant (p<0.005) drop of -2.56 p.p. in investment rates after four years. In standard deviation units, corporate investments drops by 0.19. In terms of economic magnitudes this compares to Alfaro, Bloom, and Lin (2024) who find that uncertainty shocks lead to a 0.18 standard deviation drop in investment rates.

6.2 Effects of Increased Benchmarking at the Industry-level

We use the NBER-CES Manufacturing Industry Database to estimate the effect of increases in industries' average CAPM $\widehat{\beta}$ on capital accumulation over long horizons. Our results show that higher CAPM $\widehat{\beta}$s led to 12.5% lower capital accumulation from 2000 to 2016 at the industry level.

Long-term Effects of CAPM $\widehat{\beta}$ Distortions on Capital Accumulation We use the NBER CES Manufacturing Industry database to study the long-term effects of increasing CAPM $\widehat{\beta}$s due

[29] Defined as $\frac{\text{CAPX}_t}{\frac{1}{2}(\text{PPENT}_{t-1}+\text{PPENT}_t)}$ (see e.g, Belo, Lin, and Bazdresch, 2014).

Table 3: Long-term Effects of Benchmarking on Capital Accumulation at the Industry-level

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable: log (Real Capital Stock in 2016/Real Capital Stock in 2000)						
Δ CAPM $\widehat{\beta}^A$ (2000-2016)	-0.414$^+$	-0.437*	-0.509*	-0.429*	-0.500$^+$	-0.490$^+$
	(0.243)	(0.214)	(0.218)	(0.176)	(0.276)	(0.232)
Real Capital Stock/Value Added (2000)		0.0705		-0.108		-0.0986
		(0.095)		(0.068)		(0.069)
log (Employment) (2000)		0.0290		0.0743		0.0717$^+$
		(0.037)		(0.044)		(0.039)
log (TFP) (2000)		0.427$^+$		-0.203		-0.211
		(0.221)		(0.156)		(0.154)
Pretrend Real Capital Stock (1990-1999)					-0.0771	-0.185
					(0.110)	(0.122)
Pretrend Employment (1990-1999)					0.296$^+$	0.276$^+$
					(0.147)	(0.143)
Pretrend Wages (1990-1999)					-0.0543	0.0938
					(0.446)	(0.588)
Constant	0.151$^+$	-0.0542				
	(0.082)	(0.254)				
Fixed Effects						
Subsector FE			✓	✓	✓	✓
FS F-stat.	7.54	12.93	22.31	19.54	21.42	21.85
Observations	111	111	107	107	107	107

Notes: This table reports coefficient estimates of regressions at the NAICS 5-digit industry-level of the form: $\Delta \log(\text{Real Capital Stock})_i = \alpha_j + \gamma \Delta\text{CAPM}\ \widehat{\beta}_i^A + \zeta X_i + \varepsilon_i$ in which changes in unlevered CAPM $\widehat{\beta}^A$ are instrumented with changes in BMI from 2000 to 2016. BMI and CAPM $\widehat{\beta}$ are market-value weighted averages at the NAICS 5-digit industry level of Compustat firms. Pretrends measure log changes in variables from 1990 to 1999. Regressions are weighted by industry value added in 2000. Sub-sector fixed effects are at the NAICS 3-digit level. FS F-stat is Kleibergen-Paap F-stat for weak instruments. All variables are winsorized at the 2.5% and 97.5% level. Standard errors clustered at subsector-level in parentheses. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001.

to increased benchmarking on capital accumulation at the industry-level. We estimate the long-term effects of increasing CAPM $\widehat{\beta}$s due to higher benchmarking at the NAICS-5 digit industry-level. We weigh each industry observation by its value-added in the year 2000 to obtain results comparable to the aggregate economy.

We estimate a series of IV regressions in long-differences from 2000 to 2016 of the form:

$$\Delta_{00}^{16}\text{CAPM}\ \widehat{\beta}_i^A = \delta_j + \theta\Delta_{00}^{16}\text{BMI}_i + \zeta X_i^{`00} + \epsilon_i \tag{15}$$

$$\log\left(\text{Real Capital Stock}_i^{`16}/\text{Real Capital Stock}_i^{`00}\right) = \alpha_j + \gamma\Delta_{00}^{16}\widehat{\text{CAPM}\ \widehat{\beta}_i^A} + \zeta X_i^{`00} + \varepsilon_i \tag{16}$$

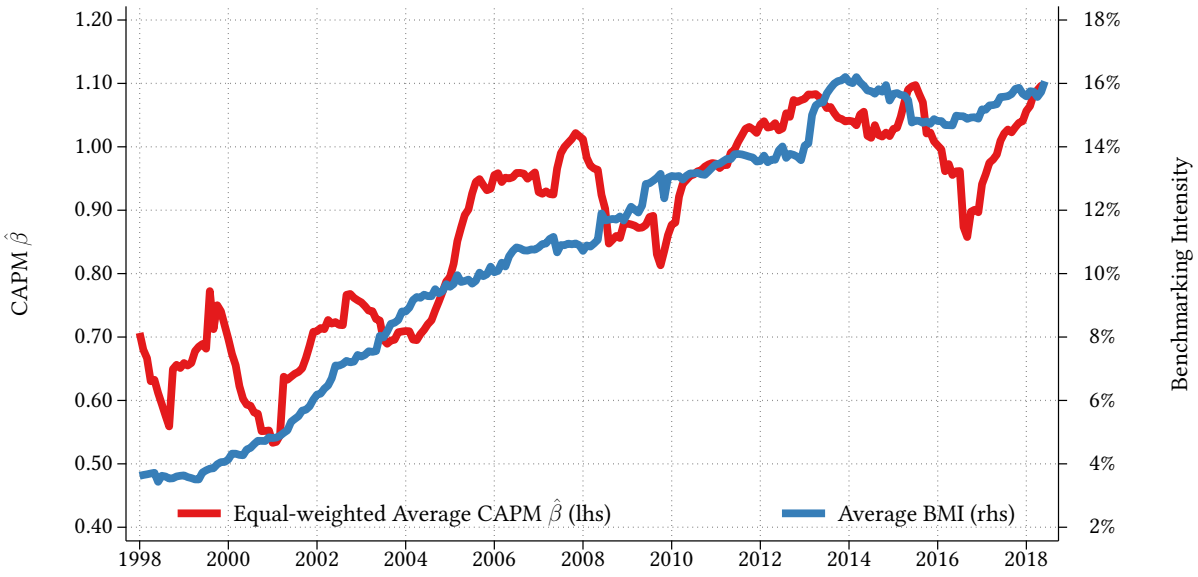
in which α_j are NAICS 3-digit subsector fixed effects. The vector of control variables $X_i^{`00}$ includes industry-level characteristics in 2000, such as the log of employment and TFP. The controls account for initial differences in industry characteristics that may affect capital accumulation. We calculate the change in CAPM $\widehat{\beta}^A$ for industry i from 2000 to 2016 as the difference between the market equity-weighted average CAPM $\widehat{\beta}^A$ of firms in that industry in 2016 and 2000. Similarly, we calculate the change in industry BMI from 2000 to 2016 as the difference between the market equity-weighted average BMI of firms in the same industry in 2016 and 2000.

A potential concern with our industry-level analysis is that the effects of increasing benchmarking on capital accumulation may be confounded by other secular changes in the economy or that industries were already on different growth paths in 2000. We address these concern by including controls for pre-trends in capital accumulation, employment, and wages from 1990 to 1999. Additionally, we include NAICS 3-digit sub-sector[30] fixed effects in the regressions. Using sub-sector fixed effects, we identify the effects of changes in CAPM $\widehat{\beta}$ on capital accumulation using variation across industries within the same sub-sectors.

Results Table 3 reports coefficient estimates of Eq. (16). Several things are worth noting. First, across all specifications, we find that increases in CAPM $\widehat{\beta}$s have a statistically significant negative effect on long-term capital accumulation at the industry-level. Second, the estimated coefficients are economically meaningful. The coefficient of Column (4) implies that the 0.29 increase in average CAPM asset $\widehat{\beta}^A$ for Russell 2000 companies (see Figure 3) is associated with a 12.5% ($\approx$0.29$\times$0.43$\times$100%) lower aggregate capital stock over the 17-year period. These magnitudes imply a user cost of capital elasticity close to the value of 1 implied by Cobb-Douglas production. Third, the results are robust to the inclusion of industry-level controls and sub-sector fixed effects. Fourth, changes in BMI are strong instruments for changes in CAPM $\widehat{\beta}$ even at the industry-level, with first-stage F-stats averaging 17.6 across columns.

[30]For example, the NAICS 3-digit sub-sector "311 - Food Manufacturing" contains 12 NAICS 5-digit industries.

Figure 7: Equal-weighted Average Cross-Sectional CAPM $\widehat{\beta}^E$ and BMI over the Past 20 Years



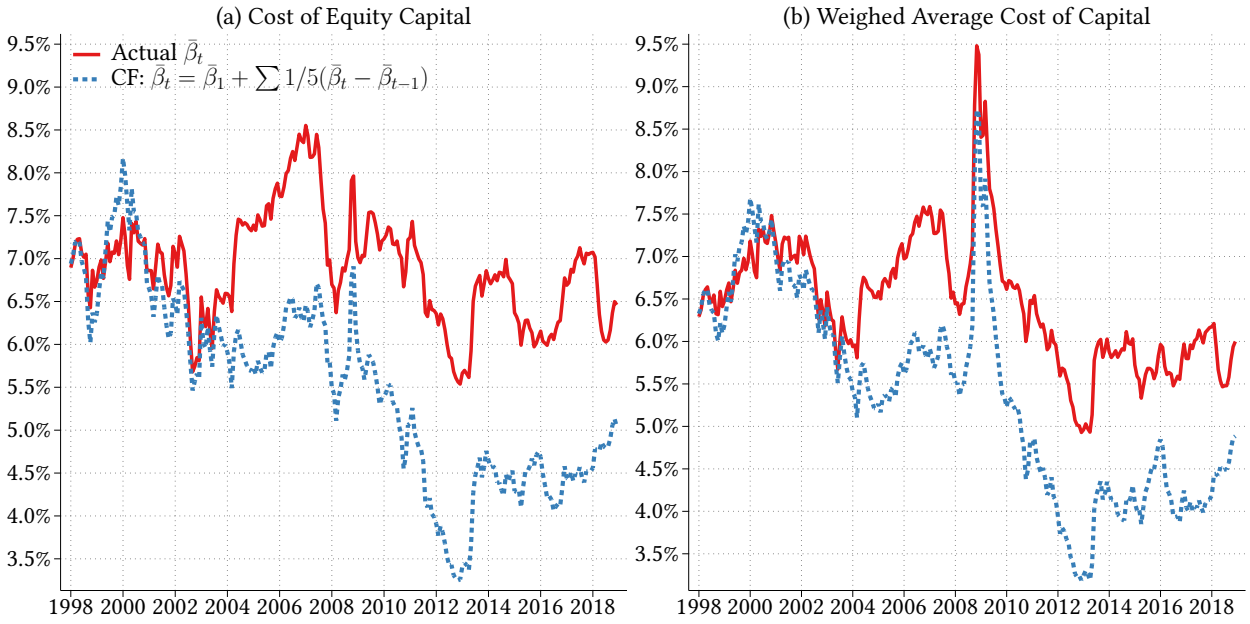
7 Effects of Increased Benchmarking in Aggregate

We argue that increased benchmarking has contributed to the missing investment puzzle documented by (Gutiérrez and Philippon, 2017). Our difference-in-differences results show that exogenous BMI increases due to benchmark inclusion raise the CAPM $\widehat{\beta}$s of firms. Extending this to the broader cross-section of firms, we estimate that the equal-weighted average of CAPM $\widehat{\beta}$s increased by 0.41 over the past 20 years after being relatively stable for 30 years.[31] We attribute almost 90% of this increase to the large increase in benchmarking intensity since 1998. We construct a counterfactual WACC which shows that the increase in benchmarking raised the average firm's WACC by 145 bps, on average. The counterfactual shows that the increase in the perceived equity risk largely offset the decline in risk-free rates over the past 20 years. The wedge is large enough to explain approximately 57% of the missing investment puzzle.

Increase in Equal-Weighted Average CAPM $\widehat{\beta}$ Over the Past 20 years Figure 7 shows the evolution of average CAPM $\widehat{\beta}$ and average BMI from 1998 to 2018. A casual inspection of the graphs suggests that both time-series are related. The average BMI increased from 4% in 1998 to 16% in 2018, while the average CAPM $\widehat{\beta}$ increased from 0.63 to 1.09.

[31] Appendix Figure A9 plots the time series of equal-weighted average CAPM $\widehat{\beta}$ from 1975 to 2020. We start in 1975 to focus on the period after NASDAQ's addition to the CRSP sample. The equal-weighted average CAPM $\widehat{\beta}$ had a mean of 0.67 from 1975 to 2000 but increased in the 2000s, settling at new mean of around 1.02 post 2010.

Figure 8: Actual and Counterfactual Estimates of Cost of Equity Capital and WACC



Notes: This figure shows monthly estimates of the cross-sectional averages of cost of equity capital and WACC. Red solid line shows cost of capital estimates using the actual measured CAPM $\widehat{\beta}_t$. Blue dashed lines shows counterfactual in which CAPM $\widehat{\beta}_t$ is adjusted for BMI increases (see Eq. 19).

We estimate how an increase in the average cross-sectional BMI affects the average CAPM $\widehat{\beta}$. To address the potential of a "spurious regression", we follow the advice of Hansen (p. 588, 2023) and include a lag of the outcome variable. We additionally report results of Dynamic OLS (DOLS) (Stock and Watson, 1993) specifications. Appendix Table A5 reports results of these time-series regressions. The results imply that a 1 p.p. increase in average BMI increases the average CAPM $\widehat{\beta}$ by 0.03 in the long-run. The rise in the average stocks's benchmarking intensity from 1998 to 2018 explains approximately 88% (=14p.p.$\times$0.029/(1.09-0.63)) of the increase in average CAPM.[32]

Counterfactual Weighted Average Cost of Capital Over the Past 20 years We next document that the benchmarking induced CAPM $\widehat{\beta}$ distortions are large enough to affect firm's weighted average cost of capital (WACC) over the past 20 years. We calculate a counterfactual WACC that adjusts the CAPM $\widehat{\beta}$ for the BMI increase and holds all other factors constant. The wedge between the actual and counterfactual WACC is on average 145 bps and is, by construction, due to the increase in CAPM $\widehat{\beta}$.

We calculate the average cross-sectional WACC in the Compustat sample as follows:

$$\overline{\text{WACC}}_t = \bar{\omega}_t \times \bar{r}_t^E + (1 - \bar{\omega}_t) \times \bar{r}_t^D \times (1 - \bar{\tau}_t) \tag{17}$$

[32]We cannot statistically reject that 100% of the increase in average CAPM $\widehat{\beta}$ is driven by the increase in BMI.

in which $\bar{\omega}_t = \bar{E}_t/(\bar{E}_t + \bar{D}_t)$ represents the fraction of firm value financed by equity (on average 0.76 over the sample) and $\bar{\tau}_t$ is the average cross-sectional tax rate (on average 0.26). The expected return on equity is given by the CAPM

$$\bar{r}_t^E = r_t^f + \bar{\hat{\beta}}_t \times \left(\mathbb{E}_t \left[r^{Mkt} \right] - r_t^f \right), \tag{18}$$

in which we set the real risk-free rate, r_t^f, to the constant maturity yield on 10-year Treasury Inflation-Protected Securities (TIPS). We assume a constant 6% ERP[33] and use the yield on the ICE-BofA High-Yield Bond Index to estimate the average expected return on debt, $\bar{r}_t^D$.[34]

We calculate a counterfactual WACC motivated by our findings that increases in BMI cause CAPM $\widehat{\beta}$s to rise and that at least 80% of the aggregate increase in CAPM $\widehat{\beta}$s was driven by increases in the BMI (see Figure 7). We adjust the average CAPM $\bar{\hat{\beta}}_t$ in Eq. (18) as:

$$\bar{\hat{\beta}}_t^{CF\ 2} = \bar{\hat{\beta}}_{1998/1} + \frac{1}{5} \times \sum_{i=1998/2}^{t} (\bar{\hat{\beta}}_i - \bar{\hat{\beta}}_{i-1}) \quad \forall\, t \geq 1998/1. \tag{19}$$

This adjustment guarantees that the counterfactual CAPM $\bar{\hat{\beta}}_t^{CF\ 2}$ is perfectly correlated with the actual CAPM $\bar{\hat{\beta}}_t$, but the increase in CAPM $\widehat{\beta}$ is exactly 20% at the end of the sample.

Figure 8 shows estimates of the actual and counterfactual cost of equity capital and WACC from 1998 to 2018. The actual cost of equity capital remained relatively stable over the time-period. The cost of equity for the average firm in 2018 is approximately 60 bps lower than in 1998. In the counterfactual scenarios, however, the cost of equity declines substantially. The secular decline in the risk-free rate over the past 20 years drives the decline in the cost of equity (Bauer and Rudebusch, 2020). The counterfactual uncovers that the decline in the risk-free rate was largely offset by an increase in equity risk premium for the average firm. The actual WACC decreased slightly over the sample period, while the counterfactuals decreased substantially. The actual WACC decreased by approximately 60 bps over the sample period. The counterfactual WACC declines by over 160 bps from 1998 to 2018.

Missing Investment in the Aggregate We assess whether the wedge between actual and counterfactual WACC, is large enough to account for the missing investment puzzle. We adopt the methods of Gutiérrez and Philippon (2017) and Gormsen and Huber (2023). Using data from

[33] Results using a time-varying ERP show that the cost of equity and WACC have increase over the past 20 years. The weighted average cost of capital would have declined had the CAPM $\widehat{\beta}$ not increased (Appendix Figure A11).

[34] The ICE-BofA yield closely tracks other common proxies for the average cost of debt. For instance, Gormsen and Huber (2024) use interest expenses over total debt in Compustat. Appendix Figure A10 shows both time-series.

1990–2002, we estimate the relationship between aggregate investment and Tobin's Q, then predict post-2002 investment under the assumption that this relationship remained unchanged. The "missing investment" is the cumulative shortfall since 2002, reflecting the divergence between Tobin's Q and observed investment. Gormsen and Huber (2023) show that when a firm's perceived discount rate exceeds the market's discount rate, it undervalues profits generated by capital relative to the market. Following their approach, we adjust Tobin's Q to account for the average discrepancy between the market's discount rate and the firm's perceived cost of capital, yielding an adjusted Tobin's Q:

$$\text{Adjusted Tobin's Q} = \text{Tobin's Q} \times \frac{1}{1 + \Delta\text{WACC} \times \text{Dur}} \tag{20}$$

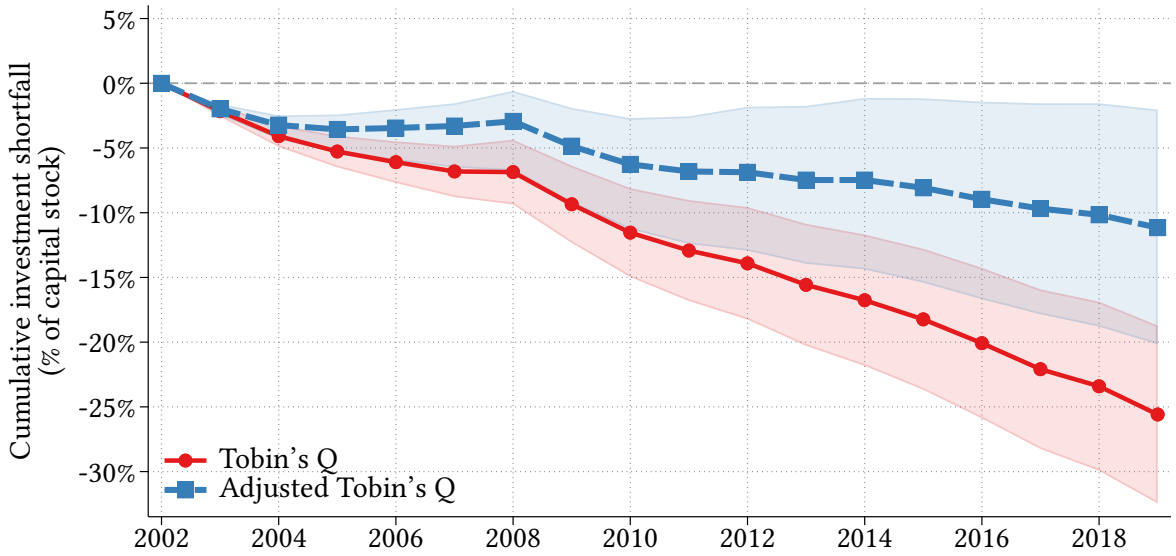
in which ΔWACC is the wedge between actual and counterfactual WACC documented in Figure 8 and Dur is cash flow duration. The impact of this adjustment depends on both the size of the discount rate wedge and the duration of the cash flows. A higher duration amplifies the effect of the discount rate on asset value. The influence of wedges thus grows with the duration of cash flows. We set the duration to 28 years which is the midpoint of the stock market duration estimates of Gormsen and Huber (2023) and Greenwald, Leombroni, Lustig, and Van Nieuwerburgh (2021).[35]

Figure 9 shows that the WACC wedge we document can explain 57% of the missing investment puzzle. Without adjustment, the aggregate investment shortfall implied by Tobin's Q is approximately 25% of the capital stock by 2019. After adjustment for the WACC wedge, the shortfall is reduced to approximately 11% of the capital stock. The wedge between actual and counterfactual WACC is thus large enough to account for more than half of the missing investment puzzle. Other channels like an increase in market power likely account for the remaining shortfall (De Loecker, Eeckhout, and Unger, 2020, Barkai, 2020, Crouzet and Eberly, 2023).

A potential concern with the aggregate results is our reliance on the equal-weighted average CAPM $\widehat{\beta}$ to calculate changes in the perceived WACC. One might argue that the market value-weighted average CAPM $\widehat{\beta}$ may be more relevant for the aggregate economy but, by definition, must always equals 1. However, its relevance for aggregate investment is less clear than one might think. Lee, Shin, and Stulz (2021) document that high market valuations for large firms are a proxy for rents rather than for investment opportunities. Jiang, Vayanos, and Zheng (2024) find that flows into passive funds raise disproportionately the stock prices of the stock market's largest firms, and especially those large firms that the market overvalues. Further, the Roll (1977) critique applies: the *stock* market value-weighted CAPM $\widehat{\beta}$ is only meaningful if stock market

[35] We obtain quantitatively and qualitatively similar results using a time-varying measures of cash flow duration.

Figure 9: Tobin's Q and the Missing Investment



Notes: This figure shows the cumulative investment shortfall as a percentage of the capital stock, estimated separately using Tobin's Q and Adjusted Tobin's Q. Following Gormsen and Huber (2023), Tobin's Q is calculated using market value data from the Flow of Funds and tangible plus intangible capital data from the BEA. Adjusted Q accounts for the wedge between financial market discount rates and firm managers' perceived cost of capital (see Eq. (20)). The relationship between investment and Q is estimated using 1990–2002 data for each Q type. For post-2002 years, cumulative residuals are computed as the difference between observed investment and predictions based on 1990–2002 estimates. Confidence intervals (95%) are derived using Newey-West standard errors with 5 lags.

values perfectly proxy aggregate investment opportunities.

We therefore explore other weighting schemes to calculate the weighted average CAPM $\widehat{\beta}$ of Compustat firms. Appendix Table A6 compares the cross-sectional weighted average CAPM $\widehat{\beta}$ of Compustat firms across two periods: 1975–2003 and 2004–2017, using seven different weighting schemes—market value, equal weight, PPENT, capital expenditures, total assets, sales, and value added. For all schemes except market value (by definition), the average CAPM $\widehat{\beta}$ increased after 2004, with changes ranging from 0.13 (value added) to 0.27 (equal weight). The increases are statistically significant at the 0.1% level for all schemes except market value.

8 Conclusion

This paper studies the causal effects of benchmarking-induced asset price distortions on corporate investment. We find that increases in benchmarking intensity cause CAPM $\widehat{\beta}$ to rise. Over the past 20 years, increased benchmarking has caused the equal-weighted average CAPM $\widehat{\beta}$ to rise by 0.41. Firms reduce investment in response to benchmarking-induced increases in CAPM $\widehat{\beta}$. We argue that this behavior results from managers' reliance on textbook guidance to esti-

mate the cost of capital using the CAPM, without internalizing the asset price distortions caused by benchmarking. Consistent with this mechanism, managers report higher perceived costs of capital in earnings calls after their CAPM $\widehat{\beta}$ increases due to benchmarking.

An influential literature shows that U.S. investment has been low relative to valuations over the past two decades. Our findings suggest that increases in average CAPM $\widehat{\beta}$ can explain up to 57% of the cumulative investment shortfall since the early 2000s. The benchmarking of asset managers and the rise in passive investing therefore have significant and economically meaningful implications for aggregate investment. Our study highlights the important role of benchmarking in shaping asset prices and corporate investment decisions. The findings underscore the need for managers, policy makers, and investors to consider the unintended consequences of the growth in benchmark-linked capital and its impact on the real economy.

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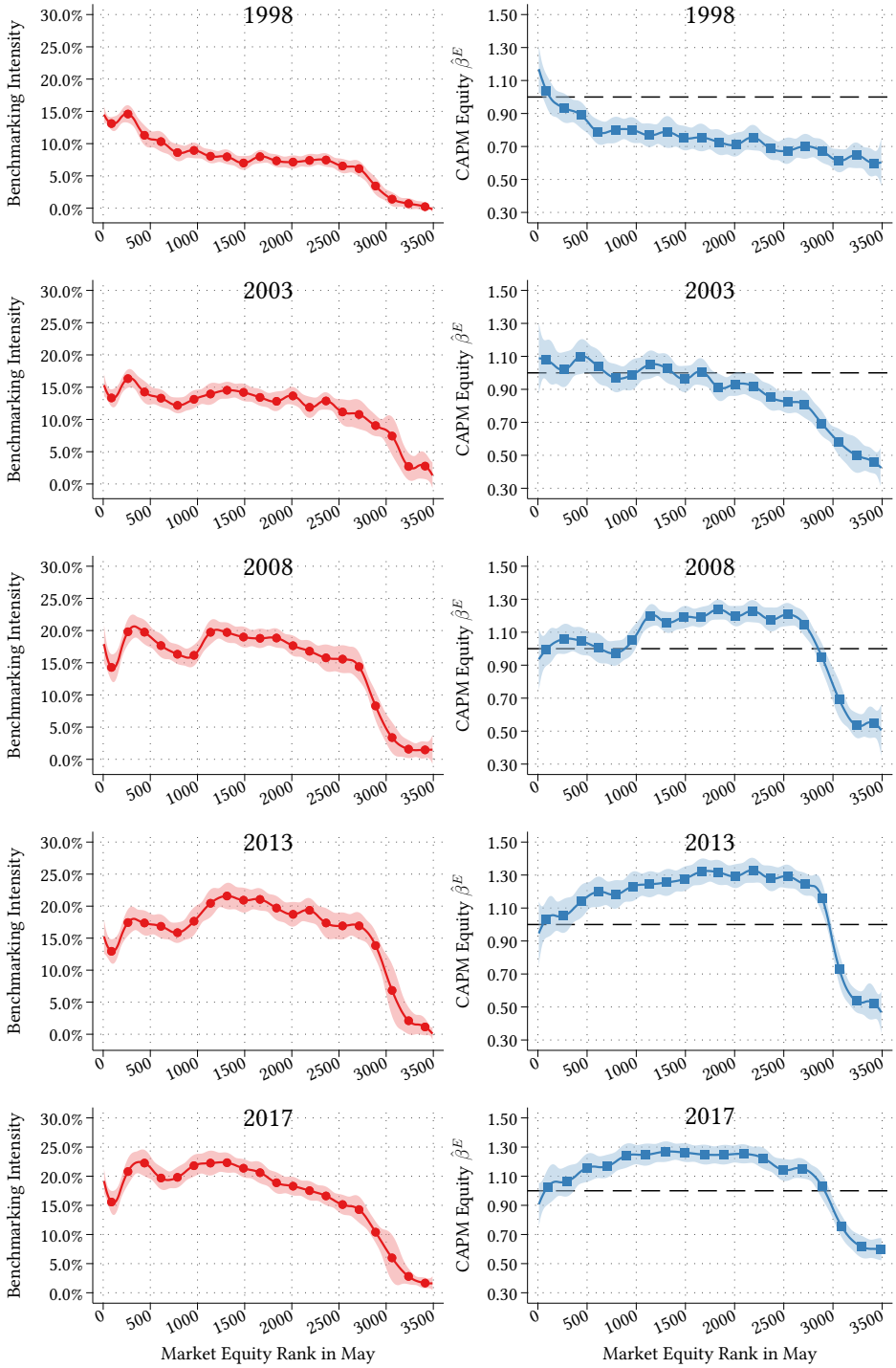
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A Appendix

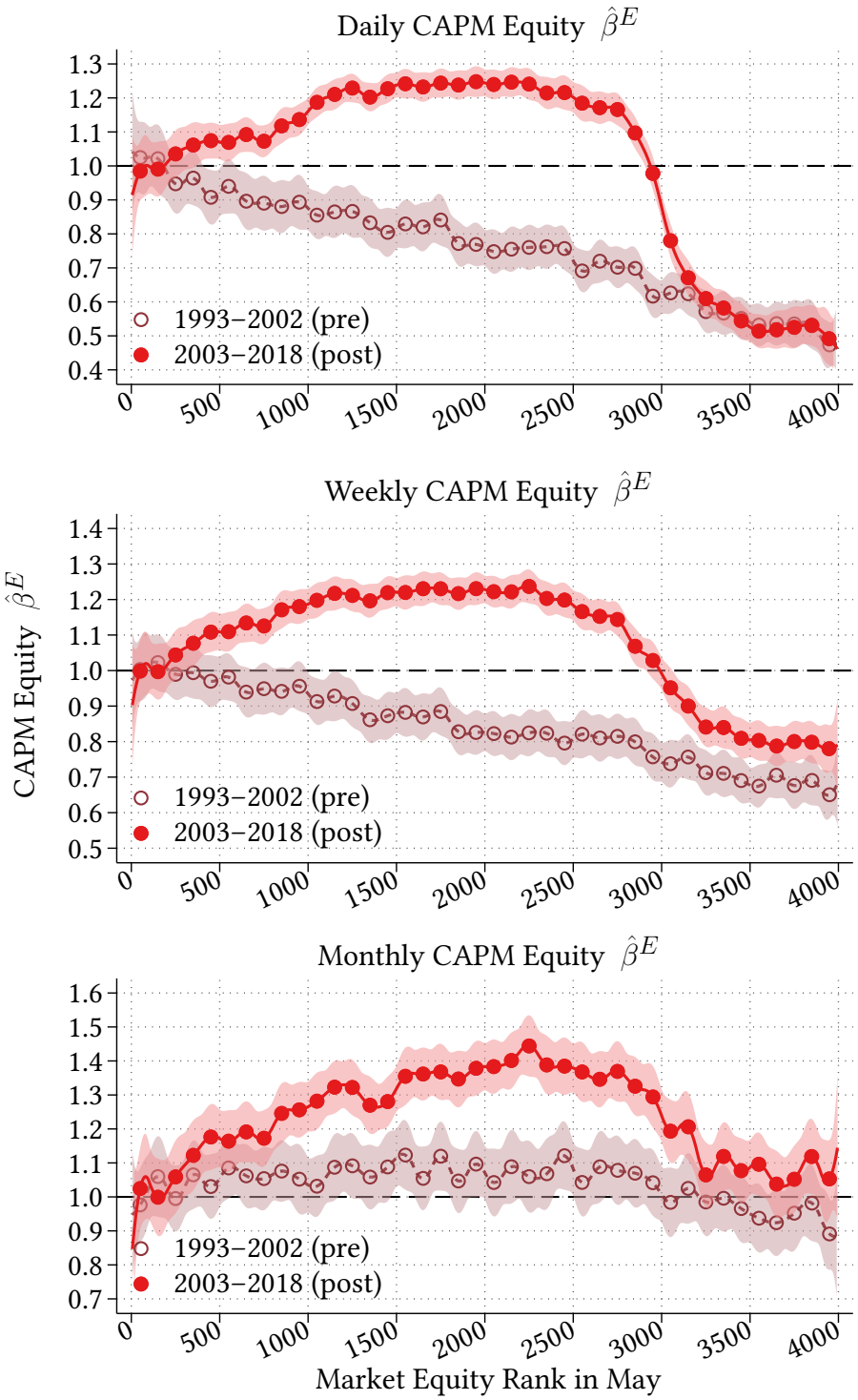
A.1 Appendix Figures

Figure A1: BMI and CAPM $\widehat{\beta}$s vs. Market Equity Rank in May over Time



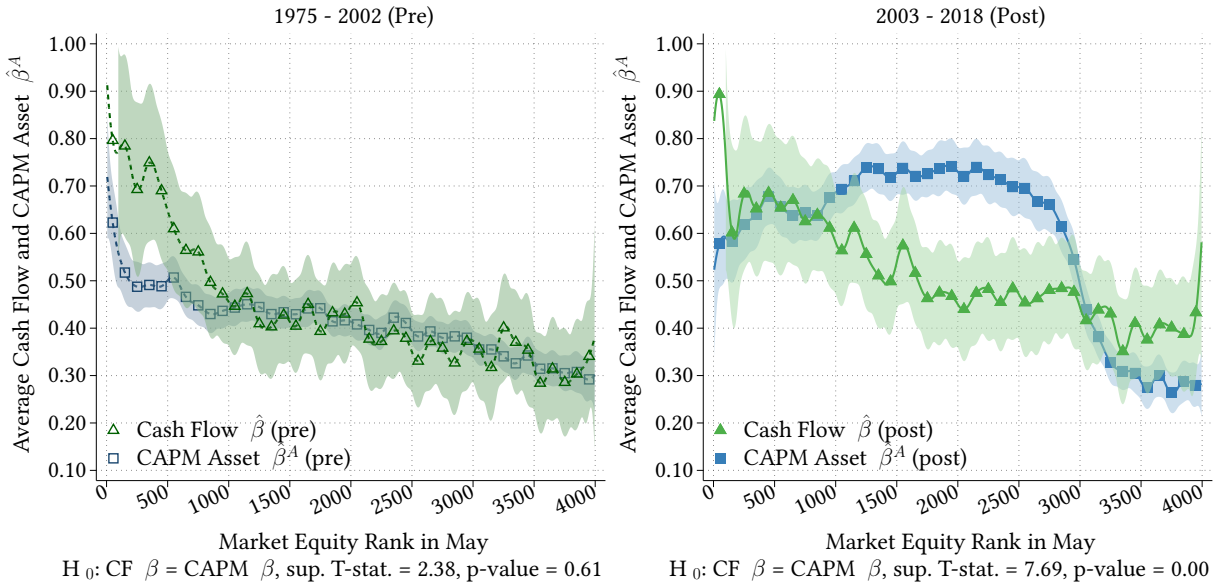
Notes: This figure shows binned scatter plots of BMI and CAPM $\widehat{\beta}$s against the market equity rank in May for the years 1998, 2003, 2008, 2013, and 2017. Shaded areas are 95% confidence bands based on standard errors clustered at the stock level.

Figure A2: Rolling-Window CAPM $\widehat{\beta}$ Estimates at Different Frequencies



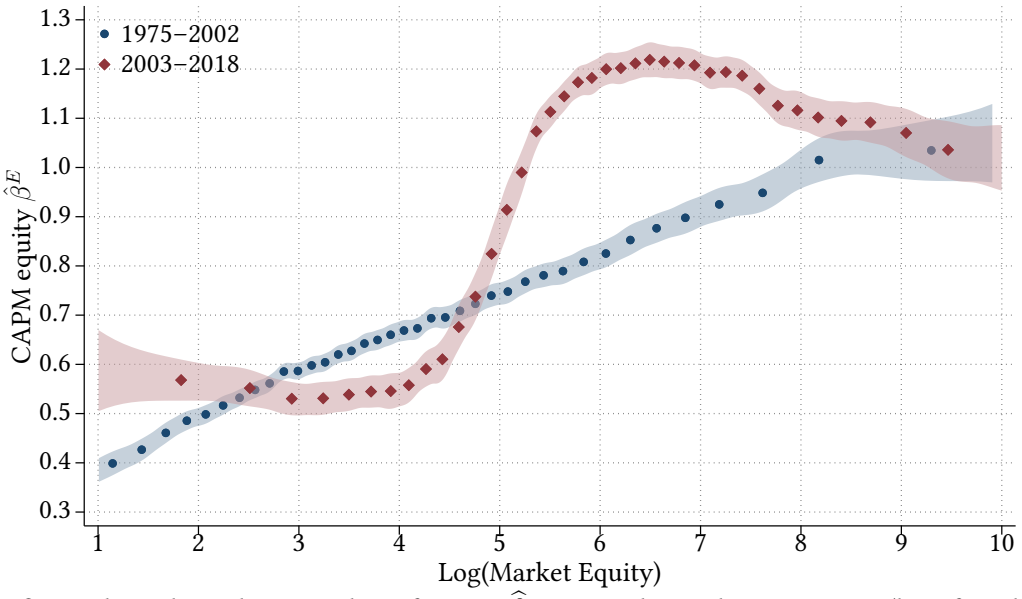
Notes: This figure shows binned scatter plots of rolling-window CAPM $\widehat{\beta}$s against the market equity rank in May for different estimation frequencies. Daily rolling window estimates use 252 trading days, weekly rolling window estimates use 156 weeks of data, and monthly rolling window estimates use 36 months of data. Shaded areas are 90% confidence bands based on standard errors clustered at the stock and year-month level.

Figure A3: Cash Flow β and CAPM Asset $\widehat{\beta}^A$ vs. Market Equity Rank in May



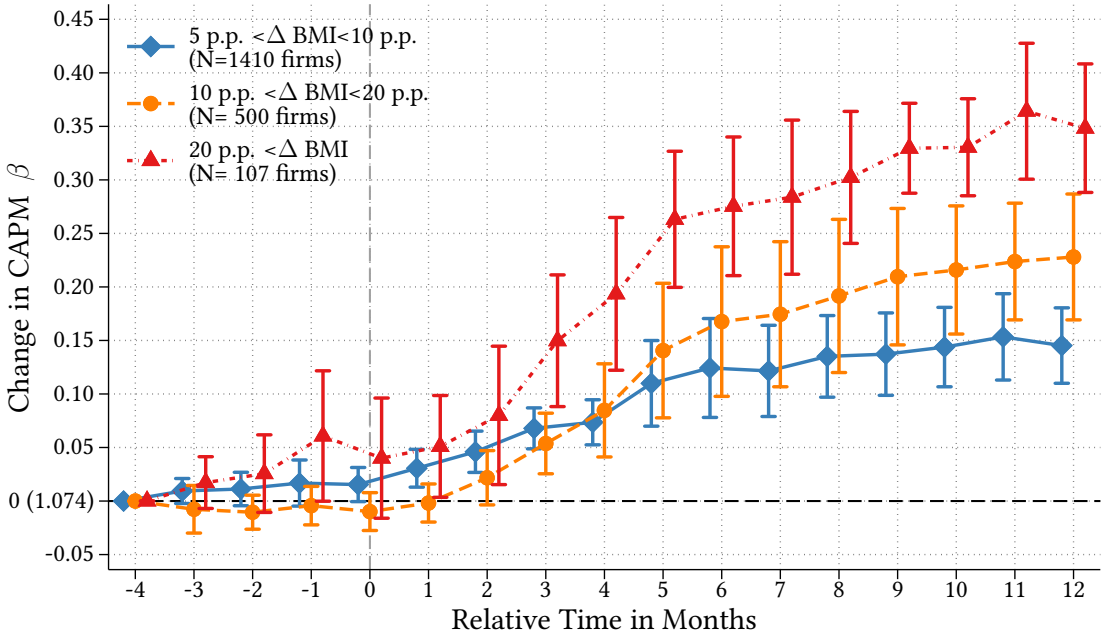
Notes: This figure shows binned scatter plots of stocks' quarterly cash flow β (in green) and CAPM asset $\widehat{\beta}$ (in blue) estimates versus their market equity ranks in May. The left panel covers 1975–2002, and the right panel covers 2003–2018. We follow Krüger et al. (2015) and unlever equity $\widehat{\beta}$s as $\beta^A = E/(E+D) \times \beta^E$ in which E is the market value of equity, D is the total book value of debt. Each bin shows the equal-weighted average of approximately 200 ranks. Conditional means across market equity ranks are identified using only cross-sectional variation by absorbing year-quarter fixed effects. Shaded areas are 95% confidence bands based on standard errors clustered at the stock and year-quarter level. Test results at the bottom are for a non-parametric pairwise comparison test (Cattaneo et al., 2024) of H_0: CF $\widehat{\beta}$ = CAPM $\widehat{\beta}$ for equity ranks 1000 to 3000.

Figure A4: Measured CAPM β^E vs. Size (Log of Market Equity)

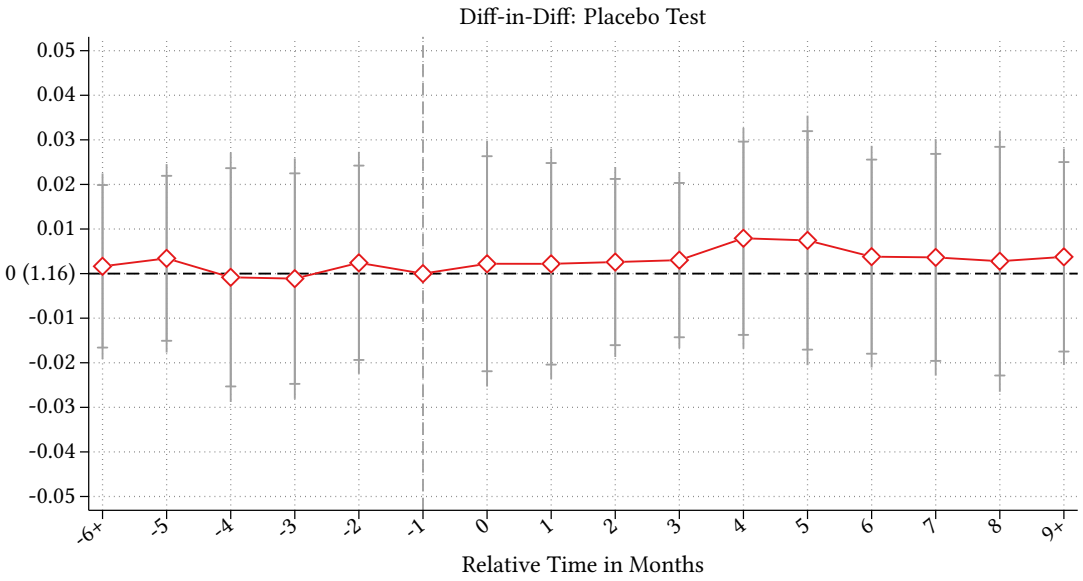


Notes: This figure shows binned scatter plots of CAPM $\widehat{\beta}$s against the market equity size (log of market equity). Conditional means across market equity ranks are identified using only cross-sectional variation by absorbing year-month fixed effects. Shaded areas are 95% confidence bands based on standard errors clustered at the stock and year-month level.

Figure A5: Difference-in-differences Event Study for Different Treatment Intensities

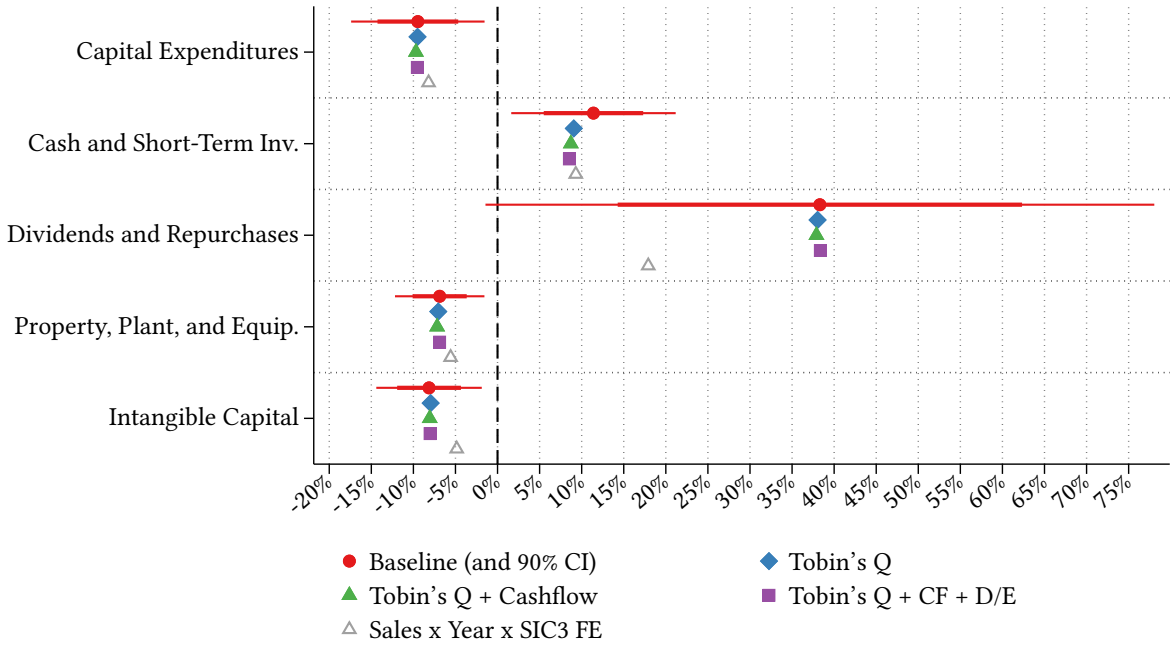


Notes: This figure shows difference-in-differences event study coefficients of Eq. (8) for different treatment intensities. Treatment effects are normalized to zero in t-4. Change in CAPM $\widehat{\beta}$ of treated stocks relative to control group in which $|\Delta$ BMI$|$<1.5 p.p.. Pointwise confidence intervals (90%) based on double-clustered standard errors. Values in parentheses on the Y-axis show the median CAPM $\widehat{\beta}$ pre-treatment for all treatment intensities.



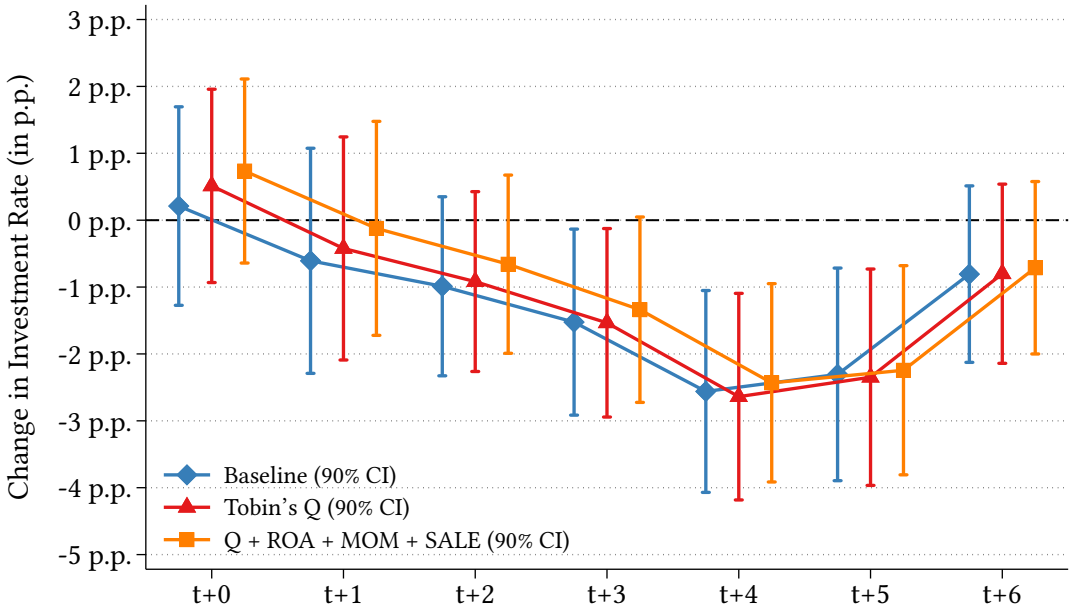
Notes: This figure shows difference-in-differences event study coefficients of Eq. (8). Treatment group: $\Delta\text{BMI} \in (0 \text{ p.p.}, 1 \text{ p.p.}]$. Control group: $\Delta\text{BMI} \in [-1 \text{ p.p.}, 0 \text{ p.p.}]$

Figure A7: LP-IV Robustness to various Alternative Specifications



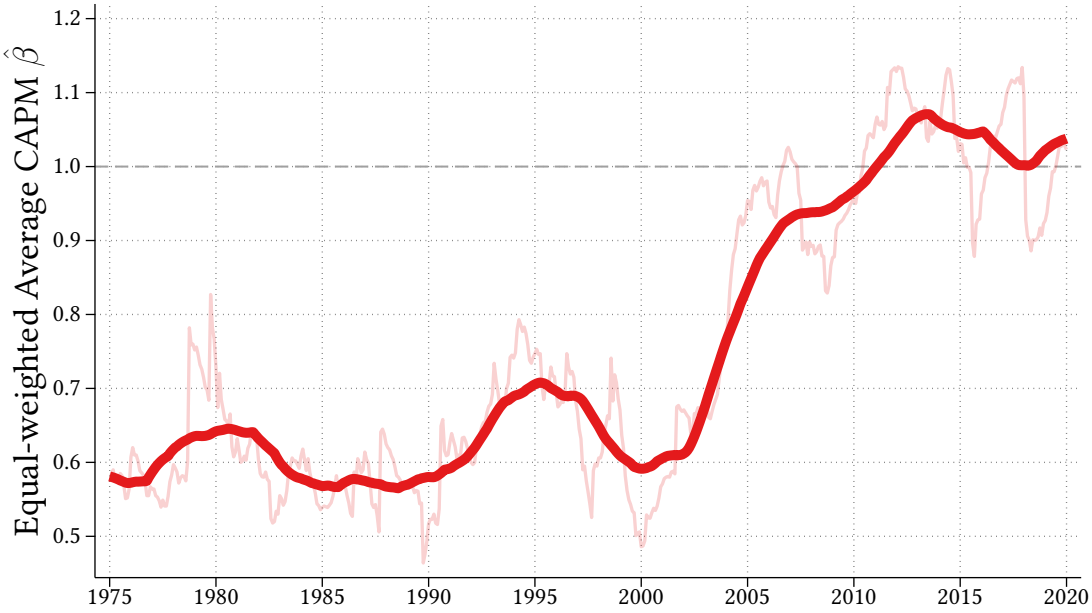
Notes: This figure shows estimates of LP-IV coefficient for h=5 from Eq. (14) for baseline and alternative specifications.

Figure A8: Impact of changes in CAPM $\widehat{\beta}$ on investment rate



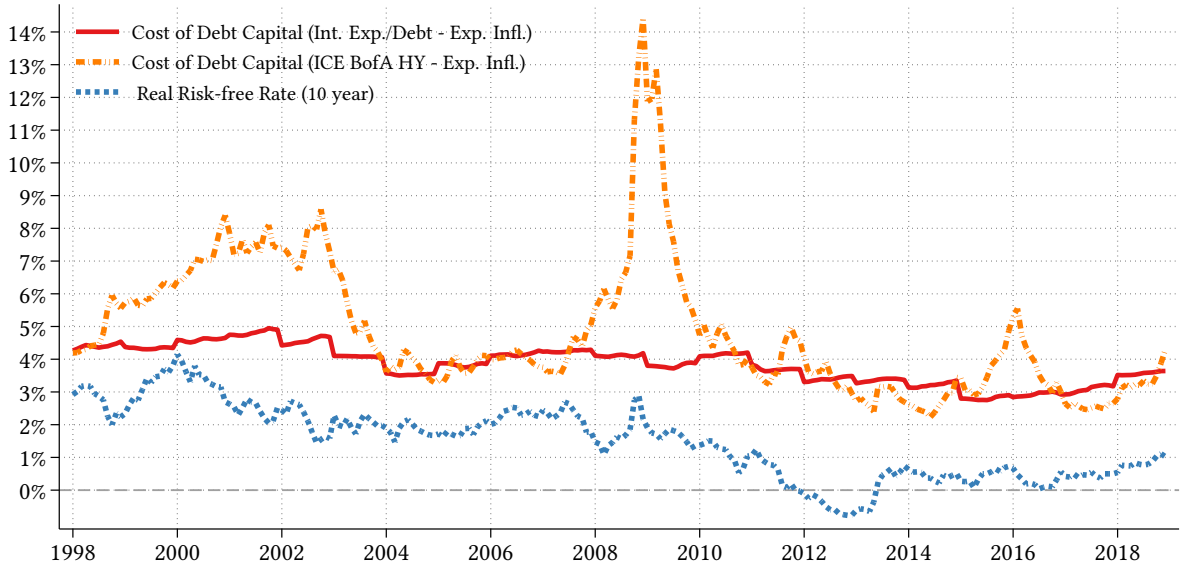
Notes: This figure shows estimates for γ^h of Investment Rate$_{i,t+h} = \alpha_i + \alpha_{j,t} + \gamma^h \widehat{\text{CAPM}}\, \beta_{i,t} + X'_{i,t}\xi + \varepsilon_{i,t+h}$, estimated using changes in BMI as an IV for changes in CAPM $\widehat{\beta}$. Estimates are scaled to a 0.16 change in CAPM $\widehat{\beta}$. Investment rate is defined as $\frac{\text{CAPX}_t}{\frac{1}{2}(\text{PPENT}_{t-1}+\text{PPENT}_t)}$ (see e.g, Alfaro et al., 2024 or Belo et al., 2014). Median and standard deviation of investment rate are 19.9% and 13.5%, respectively.

Figure A9: Time Series of Average CAPM $\widehat{\beta}$ since 1975



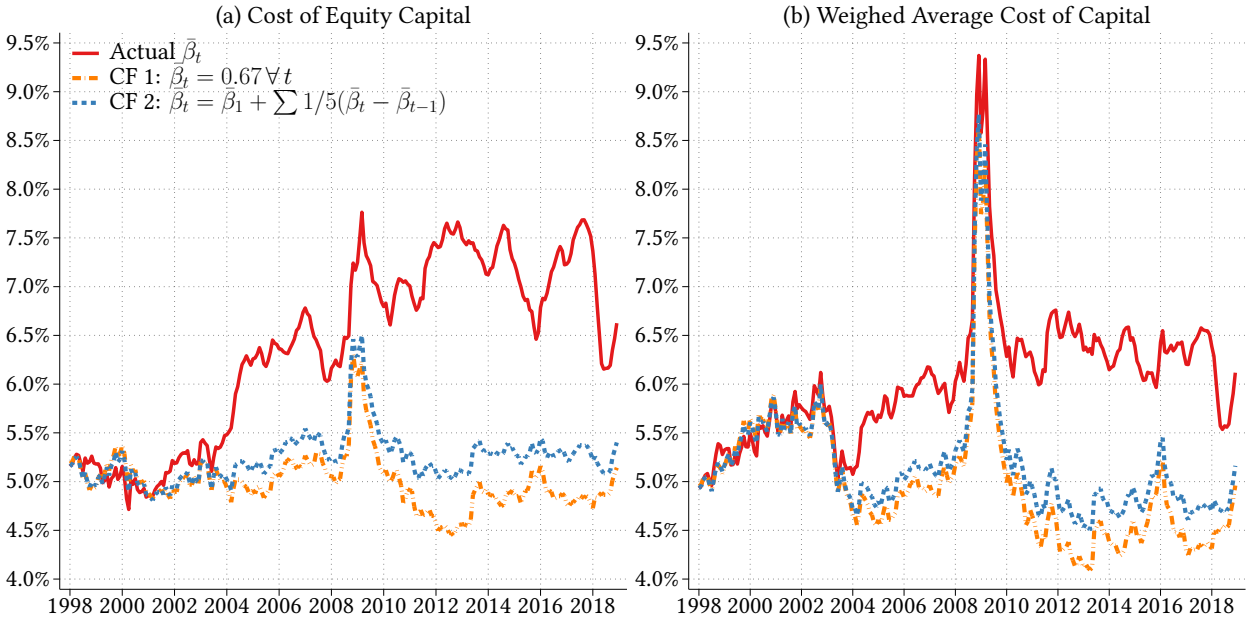
Notes: This figure shows the monthly cross-sectional average CAPM $\widehat{\beta}$ of Welch (2022b) since 1975. We focus on the period after the NASDAQ's addition to the CRSP sample. Time-series are smoothed using a two-sided moving-average filter with 24 month window on either side.

Figure A10: Average Cost of Debt Capital and Real Risk-free Rate over the Past 20 Years



Notes: This figure shows monthly estimates of the average firm's cost of debt. The solid blue line proxies for cost of debt using interest expenses over total debt in Compustat. The dashed red line proxies for cost of debt using the yield on the ICE-BofA HY index. Dashed blued line proxies for risk-free rate using the yield on 10-year TIPS from 2004 to 2018 and before 2004 nominal Treasurys adjusted for 10-year inflation expectations from the SPF.

Figure A11: Average Cost of Equity Capital and WACC over the Past 20 Years (Using Time-varying Equity Risk Premium Implied by S&P 500 Dividend Yield)



Notes: This figure shows monthly estimates of the cross-sectional averages of cost of equity capital and WACC using a time-varying ERP. We estimate $\mathbb{E}_t\left[r^{Mkt}\right] - r_t^f$ by calculating the time-varying expected return on the market (proxied by the S&P 500) using Gordon's growth model: $\mathbb{E}_t\left[r^{Mkt}\right] = D_{t+1}/P_t + \bar{g}$. We assume the average expected real dividend growth, $\bar{g}$, to be constant at 4.88%. We use the average expected dividend growth rate of the S&P 500 (7.31%) from 1994 to 2011 (Golez, 2014) and subtract expected annual inflation over the next 10 years (SPF). Red solid line shows cost of capital estimates using the actual CAPM β_t. Blue and orange dashed lines show counterfactuals in which CAPM β_t is adjusted for BMI increase (see Eq. 19) or set constant at its pre-2000 average, respectively.

A.2 Appendix Tables

Table A1: Summary Statistics of Matched BMI-CAPM Sample

	Mean	SD	Min	P5	P10	Median	P90	P95	Max	N
CAPM β (Welch, 2022)	0.88	0.48	-0.71	0.14	0.25	0.88	1.51	1.69	2.67	1,231,865
Δ CAPM β	0.02	0.27	-1.67	-0.41	-0.29	0.02	0.34	0.46	1.88	59,510
BMI in May	0.15	0.09	0.00	0.00	0.00	0.17	0.26	0.28	0.34	61,099
BMI in June	0.16	0.09	0.00	0.00	0.00	0.17	0.26	0.27	0.36	61,099
Δ BMI	0.00	0.04	-0.41	-0.04	-0.02	0.00	0.03	0.05	0.24	61,098
Market Cap. (in $m)	3,737	15,447	0.00	11.00	21.00	309.00	6,305	15,382	350,232	1,231,865
Shares Out. (in 1000s)	90.85	257.69	0.00	2.00	4.00	26.00	174.00	347.00	3,914	1,231,865
Trading Vol. (in 100,000s)	16,842	50,099	0.00	29.00	73.00	2,630	38,506	75,318	1,085,943	1,231,865

Notes: Monthly sample from 1998 to 2019. ΔCAPM $\widehat{\beta}$ is the difference between the average CAPM $\widehat{\beta}$ in the first and last quarter of a year. Variables are winsorized at the 0.5% and 99.5% level.

Table A2: Effect of Change in Benchmarking Intensity on CAPM $\widehat{\beta}$

Treatment Group:	Δ BMI>5p.p.							
Sample:	Market Equity $\geq$ P10 NYSE				Market Equity $\geq$ P20 NYSE			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Treated $\times$ Post	0.111***	0.117***	0.111***	0.111***	0.102***	0.109***	0.102***	0.103***
	(0.013)	(0.012)	(0.012)	(0.012)	(0.014)	(0.012)	(0.013)	(0.013)
Controls	✓	✓	✓	✓	✓	✓	✓	✓
Fixed Effects								
Firm $\times$ Cohort	✓	✓	✓	✓	✓	✓		
Time $\times$ Cohort	✓				✓			
Size Decile $\times$ Time $\times$ Cohort		✓				✓		
Volume Decile $\times$ Time $\times$ Cohort			✓				✓	
Shrs. Out. Decile $\times$ Time $\times$ Cohort				✓				✓
Observations	291,822	291,754	291,688	291,617	249,916	249,840	249,745	249,703

Notes: This table reports coefficient estimates of Eq (8). Treated$\times$Post is the average of the post-treatment coefficients after 5 months (to account for the expanding-window estimation of CAPM $\widehat{\beta}$). Sample from 2000-01 to 2019-06. Standard errors in parentheses are double-clustered at firm and year-month level. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001.

Table A3: Robustness to Alternative CAPM $\widehat{\beta}$ Estimators

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Panel A: CAPM β estimates using expanding windows of daily data with exponentially decaying weights of 3-months half life						
	Δ CAPM β^{OLS}_{EW}	Δ CAPM β^{WEL}_{EW}	Δ CAPM β^{DIM}_{EW}	Δ CAPM β^{BLU}_{EW}	Δ CAPM β^{TOP}_{EW}	$\Delta\ \rho(r_i, r_m)_{EW}$	$\Delta\ \sigma^i_{EW}$
Δ BMI (in p.p.)	0.0183***	0.0154***	0.0135***	0.0122***	0.0171***	0.00646***	-0.0000668*
	(0.001)	(0.001)	(0.002)	(0.001)	(0.001)	(0.000)	(0.000)
Firm FE	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓
Adj. R^2	0.15	0.16	0.11	0.15	0.26	0.41	0.59
Observations	28,514	28,514	28,514	28,514	28,514	28,514	28,514
	Panel B: CAPM β estimates using daily data with a 2-year rolling window (with equal weights)						
	Δ CAPM β^{OLS}_{RW}	Δ CAPM β^{WEL}_{RW}	Δ CAPM β^{DIM}_{RW}	Δ CAPM β^{BLU}_{RW}	Δ CAPM β^{TOP}_{RW}	$\Delta\ \rho(r_i, r_m)_{RW}$	$\Delta\ \sigma^i_{RW}$
Δ BMI (in p.p.)	0.00630***	0.00547***	0.00576***	0.00420***	0.00691***	0.00271***	-0.0000516**
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.000)	(0.000)
Firm FE	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓
Adj. R^2	0.21	0.22	0.17	0.21	0.31	0.43	0.58
Observations	28,514	28,514	28,514	28,514	28,514	28,514	28,514

Notes: This table reports coefficient estimates of specifications of the form: $\Delta\beta_{i,t} = \alpha_i + \alpha_t + \gamma\Delta\text{BMI}_{i,t} + \varepsilon_{i,t}$. $\Delta\beta_{i,t}$ is between 1st and 4th quarter of each year. β^{WEL} is estimator of Welch (2022b), β^{DIM} is estimator of Dimson (1979), β^{BLU} is estimator of Blume (1975) (also known as Bloomberg $\widehat{\beta}$), β^{TOP} is $\widehat{\beta}$ with respect to ten largest stocks by market capitalization, EW stands for expanding window, RW for rolling window. Changes in BMI and CAPM $\widehat{\beta}$s are winsorized at the 2% and 98% level. Estimation sample is restricted to stocks within 250 ranks around Russell index cutoffs. Standard errors in parentheses are clustered at firm-level. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001.

Table A4: Lag-selection based on Bayesian/Schwartz Information Criterion

Maximum lag length	L=0	L=1	L=2	L=3	L=4	L=5	L=6
Bayesian Information Criterion (OLS)	451	391	389	310	187	**157**	165
Bayesian Information Criterion (IV)	1,970	755	518	460	**411**	577	628
Observations	14,495	14,495	14,495	14,495	14,495	14,495	14,495

Notes: This table reports BIC for regression specifications of the form: ΔPerceived Cost of Capital$_{i,t} = \alpha_i + \alpha_t + \gamma\ \Delta\text{CAPM}\ \beta_{i,t} + \nu_{i,t}$ for OLS and IV regression specification in which changes in CAPM $\widehat{\beta}$ are instrumented by changes in BMI.

Table A5: Effects of Benchmarking Intensity on Equal-Weighted Average CAPM $\widehat{\beta}^E$

	(1) OLS	(2) ADL(1)	(3) DOLS(3)	(4) DOLS(4)	(5) DOLS(5)	(6) DOLS(7)
BMI (in %)	0.038	0.030*	0.032***	0.031***	0.031***	0.029***
		(0.015)	(0.005)	(0.005)	(0.005)	(0.005)
Engle-Granger's Augmented Dickey-Fuller test (H_0: no cointegration)		-13.77***	-3.93***	-3.66**	-3.75**	-3.30*
Adj. R^2		0.99	0.71	0.72	0.74	0.78
Observations	225	225	225	225	225	225

Notes: Newey-West standard error in parentheses with L=21 lags. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001.

Table A6: Weighted Average CAPM $\widehat{\beta}$ of Compustat Firms from 1975 to 2017

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
			Weighted CAPM β Estimates				
weighted by:	Market Cap.	Equal	PPENT	Cap. Ex.	Total Assets	Sales	Value Add.
$\mathbb{1}\{\text{Year} > 2003\}$	0.003	0.273***	0.196***	0.152***	0.216***	0.163***	0.130***
	(0.012)	(0.021)	(0.030)	(0.027)	(0.024)	(0.028)	(0.025)
Constant	1.001***	0.735***	0.854***	0.907***	0.849***	0.910***	0.927***
	(0.010)	(0.012)	(0.032)	(0.028)	(0.020)	(0.029)	(0.025)
Observations	43	43	43	43	43	43	43

Notes: This table reports coefficient estimates of yearly regressions of the form: Weighted CAPM $\beta_t = \alpha_0 + \mathbb{1}\{\text{Year} > 2003\} + \varepsilon_t$ from 1975 to 2017 in which Weighted CAPM $\beta_t = \sum_i \omega_i \times$ CAPM $\beta_{i,t}$ is the cross-sectional weighted average of Compustat firm's CAPM $\widehat{\beta}$. Firm-level CAPM $\widehat{\beta}$s are estimates from Welch (2022b). Newey-West standard errors with $8 = \lfloor 1.3\sqrt{43} \rfloor$ lags in parenthesis. . + p<0.10, * p<0.05, ** p<0.01, *** p<0.001.

B Additional Tests and Instrument Validity

B.1 Changes in BMI and Measures of Risk Exposure

Changes in BMI that correlate with changes in exposure to aggregate or idiosyncratic risk pose a threat to our identification strategy. Industry's exposure to aggregate risk (Karolyi, 1992) and firm fundamentals (Gomes, Kogan, and Zhang, 2003) determine firm-level exposure to aggregate risk. We test whether the aggregate risk exposure of treated firms changes by estimating whether the CAPM $\widehat{\beta}$ of comparable peer firms changes when a firm's BMI changes. We also test whether measures of idiosyncratic firm-level risk exposure change with BMI. However, we find no evidence that changes in BMI correlate with changes in risk exposure.

Changes in CAPM $\widehat{\beta}$s of peer firms We collect information about a firm's peer group from ISS. For each firm, we randomly select three peer firms and test whether the firm's change in BMI correlates with changes in the CAPM $\widehat{\beta}$ of peers. To avoid confounding our estimates, we exclude peers that also experience a change in BMI. Appendix Table A7 shows the results of this test using a firm's peers. The regression of changes in a firm's CAPM $\widehat{\beta}$ on changes in its peers' CAPM $\widehat{\beta}$ shows a significant positive coefficient, indicating common exposure to aggregate risk.[36] However, changes in a firm's BMI do not correlate with changes in peers' CAPM $\widehat{\beta}$s, with insignificant coefficients close to zero. This suggests that benchmarking distortions cause the significant changes in a firm's CAPM $\widehat{\beta}$ rather than changes in aggregate risk exposure.

[36] Levi and Welch (2017) similarly show that the CAPM $\widehat{\beta}$s of peer firms predict own firms' $\widehat{\beta}$ well.

Table A7: Placebo test using the CAPM $\widehat{\beta}$ of peer firms

	(1) Firm's Δ CAPM β^E	(2) Firm's Δ CAPM β^E	(3) All Peers' Δ CAPM β^E	(4) Peer n=1's Δ CAPM β^E	(5) Peer n=2 Δ CAPM β^E	(6) Peer n=3's Δ CAPM β^E
Δ CAPM β^{Peer}	0.232***					
	(0.010)					
Δ BMI		0.814***	0.037	0.075	-0.009	0.043
		(0.054)	(0.029)	(0.046)	(0.043)	(0.045)
Fixed Effects						
Firm FE	✓	✓	✓	✓	✓	✓
Peer FE	✓		✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Adj. R^2	0.421	0.249	0.472	0.466	0.471	0.479
Observations	46,689	16,995	47,749	16,470	16,022	15,257

Notes: This table reports coefficient estimates for a placebo test using N=3 firm peers' change in CAPM $\widehat{\beta}$ and assigns them the Δ BMI of the firm: $\Delta\text{CAPM } \beta^{\text{Peer}}_{j,t} = \alpha_i + \alpha_j + \alpha_t + \Delta\text{BMI}^{\text{Firm}}_{i,t} + \varepsilon_{j,i,t}$ for firm i and peer j in year t. Standard errors in parentheses are clustered at the firm-level in column (2) and double-clustered at firm and peer level in other columns. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001.

Firm-level risk measures We analyze six firm-level risk measures derived from earnings calls by Hassan et al. (2019): the overall risk exposure of firms, exposure to overall political risk, and exposure to political risk stemming from economic policy, security policy, technological policy, and trade policy. Appendix Table A8 reports estimates of OLS regressions of changes in firm-level risk measures on changes in CAPM $\widehat{\beta}$ and changes in BMI. Two things are worth noting. First, changes in the CAPM $\widehat{\beta}$ correlate with changes in the firm-level risk measures. Four of six firm-level risk-measure show a statistically significant positive relationship with changes in the CAPM $\widehat{\beta}$ of firms. Second, changes in BMI do not correlate with changes in firm-level risk measures. The estimated coefficients across all risk measures are close to zero and not statistically significant.

B.2 Changes in BMI and Measures of Financial Constraints

Changes in BMI could correlate with changes in financial constraints, potentially violating the exclusion restriction of our IV strategy. We test this by examining the correlation between changes in a firm's BMI and measures of financial constraints and CDS spreads. If changes in BMI correlated with changes in financing costs due to factors other than CAPM $\widehat{\beta}$, the exclusion restriction would be violated. However, we find no evidence of such correlations.

Table A8: Changes in CAPM $\widehat{\beta}$ and firm-level risk measures of Hassan et al. (2019)

(in σ units)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Δ Risk		Δ Pol. Risk		Δ Pol. Risk - Econ.		Δ Pol. Risk - Secu.		Δ Pol. Risk - Tech.		Δ Pol. Risk - Trade	
Δ CAPM β^E	0.0192**		0.0170*		0.0163*		0.0149*		0.0087		-0.0002	
	(0.007)		(0.007)		(0.007)		(0.007)		(0.007)		(0.007)	
Δ BMI		-0.0086		0.0080		-0.0011		0.0097		0.0115		-0.0088
		(0.009)		(0.009)		(0.009)		(0.009)		(0.009)		(0.009)
Fixed Effects												
Firm FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Adj. R^2	0.14	0.14	0.13	0.13	0.13	0.13	0.13	0.13	0.12	0.12	0.13	0.13
Observations	29,970	29,970	29,985	29,985	29,963	29,963	29,978	29,978	29,982	29,982	29,976	29,976

Notes: This table reports coefficients estimates for regression specifications of the form: $\Delta\text{Firm-level Risk}_{i,t} = \alpha_i + \alpha_t + \gamma\, \Delta\text{BMI}_{i,t} + \nu_{i,t}$. Changes in firm-level risk (Hassan et al., 2019) calculated between 1st and 4th quarter of the year. Coefficients are standardized to unit variances. Changes in firm-level risk measures, CAPM $\widehat{\beta}$s, and BMI are trimmed at the 1% and 99% level. Standard error in parentheses are clustered at the firm-level. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001.

Table A9: Changes in measures of text-based financial frictions (Hoberg and Maksimovic, 2015)

(in σ units)	(1) Δ Inv. Delay	(2) Δ Inv. Delay & Equity Issue	(3) Δ Inv. Delay & Debt Issue	(4) Δ Inv. Delay & Private Issue	(5) Δ Inv. Delay & Equity (LW, '23)	(6) Δ Inv. Delay & Debt (LW, '23)
Δ BMI	-0.0008	-0.007	-0.0004	-0.005	-0.010	0.004
	(0.009)	(0.009)	(0.009)	(0.009)	(0.008)	(0.007)
Fixed Effects						
Firm FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Adj. R^2	0.08	0.07	0.06	0.07	0.07	0.04
Observations	23,463	23,463	23,463	23,463	32,275	32,275

Notes: This table reports coefficients estimates for regression specifications of the form: $\Delta\text{Measure of Financial Constraint}_{i,t} = \alpha_i + \alpha_t + \gamma\, \Delta\text{BMI}_{i,t} + \nu_{i,t}$. Changes in text-based financial constraint measures from Hoberg and Maksimovic (2015) and Linn and Weagley (2021). Coefficients are standardized to unit variances. Changes in financial constraints measures, CAPM $\widehat{\beta}$s, and BMI are trimmed at the 1% and 99% level. Standard error in parentheses are clustered at the firm-level. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001.

Text-based measures of financial constraints We collect text-based measures of financial constraints from Hoberg and Maksimovic (2015) and Linn and Weagley (2021). These measures capture the extent to which firms face financial constraints and are likely to constrain investment based on the text of their annual reports. Appendix Table A9 reports estimates of OLS regressions of changes in the firm's financial constraints on changes in the BMI of a firm. The estimated coefficients of BMI are close to zero and not statistically significant across all measures. Importantly, Column (1) of Appendix Table A9 shows that changes in BMI do not correlate with firm statements about plans to delay investments.

Changes in CDS spreads and CDS CAPM $\widehat{\beta}$ We collect CDS spreads for senior unsecured debt with tenor of 5 year from 2010[37] to 2019. We calculate CDS CAPM $\widehat{\beta}$s on daily data using the estimator of Welch (2022b). We calculate changes in a firm's CDS spreads and firm's CAPM $\widehat{\beta}$ of CDS spreads as the difference between the average of daily observations in the first and last

[37]We focus on the period after ISDA's "Big Bang" reforms of April 2009 to maintain a consistent sample.

Table A10: Changes in CDS Spreads and CAPM $\widehat{\beta}$s of CDS Spreads

Dependent variable:	Δ CDS Spread (in σ units)				Δ CDS CAPM β (in σ units)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Δ BMI (in σ units)	-0.0221	-0.0244	-0.0189	-0.0283	0.0280	0.0189	0.0014	0.0305
	(0.0212)	(0.0209)	(0.0207)	(0.0210)	(0.0240)	(0.0242)	(0.0253)	(0.0260)
Momentum (Cum. Ret.)		-0.120***	-0.140***			-0.0618**	-0.0366	
(in σ units)		(0.0315)	(0.0332)			(0.0238)	(0.0259)	
Fixed Effects								
Firm FE	✓	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓			✓	✓		
Year $\times$ Size Decile			✓				✓	
Year $\times$ Momentum Decile				✓				✓
Adj. R^2	0.260	0.269	0.303	0.315	0.103	0.106	0.179	0.155
Observations	2,798	2,798	2,798	2,798	2,299	2,299	2,299	2,299

Notes: This table reports coefficients estimates for regression specifications of the form: $\Delta\text{CDS Spreads}_{i,t} = \alpha_i + \alpha_t + \gamma\, \Delta\text{BMI}_{i,t} + \nu_{i,t}$. Coefficients are standardized to unit variances. CDS spreads for senior unsecured debt with tenor of 5 year and doc clause XR14 (no restructuring). CDS CAPM $\widehat{\beta}$s are calculated on daily data from 2010 to 2019 using the weighted least squares estimator of Welch (2022b) with exponentially decay of 3 months half life. Changes in CDS spreads and CDS CAPM $\widehat{\beta}$s are trimmed at the 2% and 98% level. Standard error in parentheses are clustered at the firm-level. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001.

quarter of a year. Appendix Table A10 reports estimates of OLS regressions of changes in CDS spreads and changes in the CDS CAPM $\widehat{\beta}$ on changes in the BMI of a firm. We find no evidence that changes in the BMI predict changes in firm CDS spreads or CDS CAPM $\widehat{\beta}$s. The estimated coefficients on BMI are insignificant and close to zero.

B.3 Changes in BMI and Measures of Corporate Governance

An increase in BMI and associated institutional ownership could impact investment through improved corporate governance (Appel et al., 2016, Aghion, Van Reenen, and Zingales, 2013). However, increased passive ownership may also decrease monitoring incentives, as in the model of Bebchuk and Hirst (2019). We test whether measures of governance change with changes in BMI but find no evidence of such an effect.

We obtain governance and ESG scores from S&P, Sustainalytics, and Refinitiv and test whether changes in BMI correlate with changes in those scores. Appendix Table A11 reports estimates of OLS regressions of changes in governance and ESG scores on changes in the BMI of a firm. The estimated coefficients are close to zero and are not statistically significant. Our findings are consistent with Kacperczyk et al. (2021), who also find no evidence of changes in governance at benchmark inclusion.

Table A11: Changes in measures of corporate governance

(in σ units)	(1) Δ S&P G-Score	(2) Δ Sus. G-Score	(3) Δ Ref. G-Score	(4) Δ Sus. ESG	(5) Δ S&P ESG	(6) Δ Ref. ESG
Δ BMI	-0.024	-0.020	0.008	-0.009	0.031	-0.003
	(0.057)	(0.024)	(0.017)	(0.026)	(0.057)	(0.017)
Fixed Effects						
Firm FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Adj. R^2	0.31	0.20	0.07	0.23	0.34	0.10
Observations	2,003	7,168	13,925	7,326	2,003	13,925

Notes: This table reports coefficients estimates for regression specifications of the form: $\Delta\text{Governance Score}_{i,t} = \alpha_i + \alpha_t + \gamma\ \Delta\text{BMI}_{i,t} + \nu_{i,t}$. Governance and ESG scores of Standard & Poor, Sustainalytics, and Refinitiv. Coefficients are standardized to unit variances. Changes in G-Scores, ESG Scores, and BMI are trimmed at the 1% and 99% level. Standard error in parentheses are clustered at the firm-level. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001.

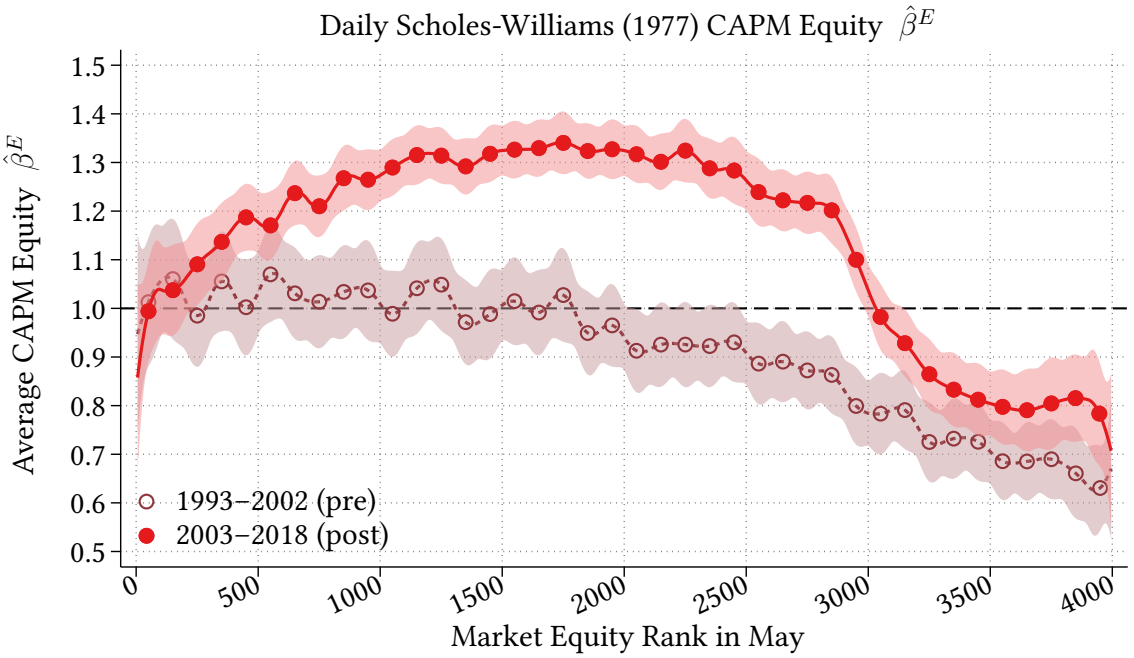
C Online Appendix (for online publication only)

Table B1: Robustness to Industry CAPM $\widehat{\beta}$ Estimators (scaled to 10 p.p. increase in BMI)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Δ CAPM β^{OLS}_{RW}		Δ CAPM β^{WEL}_{RW}		Δ CAPM β^{DIM}_{RW}		Δ CAPM β^{BLU}_{RW}	
Δ BMI	0.193*	0.169**	0.157*	0.140**	0.281**	0.221***	0.129*	0.113**
(scaled to 10 p.p. incr.)	(0.071)	(0.051)	(0.060)	(0.043)	(0.085)	(0.060)	(0.047)	(0.034)
Industry Classification	FF30	FF48	FF30	FF48	FF30	FF48	FF30	FF48
Industry FE	✓	✓	✓	✓	✓	✓	✓	✓
Adj. R^2	0.03	0.03	0.03	0.04	0.05	0.04	0.03	0.03
Observations	507	753	507	753	507	753	507	753

Notes: This table reports coefficient estimates of specifications of the form: $\Delta\beta_{i,t} = \alpha_i + \gamma\Delta\text{BMI}_{i,t} + \varepsilon_{i,t}$ at the industry level i for Fama-French's 30 and 48 industries. Estimates are scaled to a 10 p.p. increase in BMI at the industry level. β^{WEL} is estimator of Welch (2022b), β^{DIM} is estimator of Dimson (1979), β^{BLU} is estimator of Blume (1975) (also known as Bloomberg $\widehat{\beta}$), RW stands for rolling window. Changes in BMI and CAPM $\widehat{\beta}$s are winsorized at the 1% and 99% level. Standard errors in parentheses are clustered at industry-level. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001.

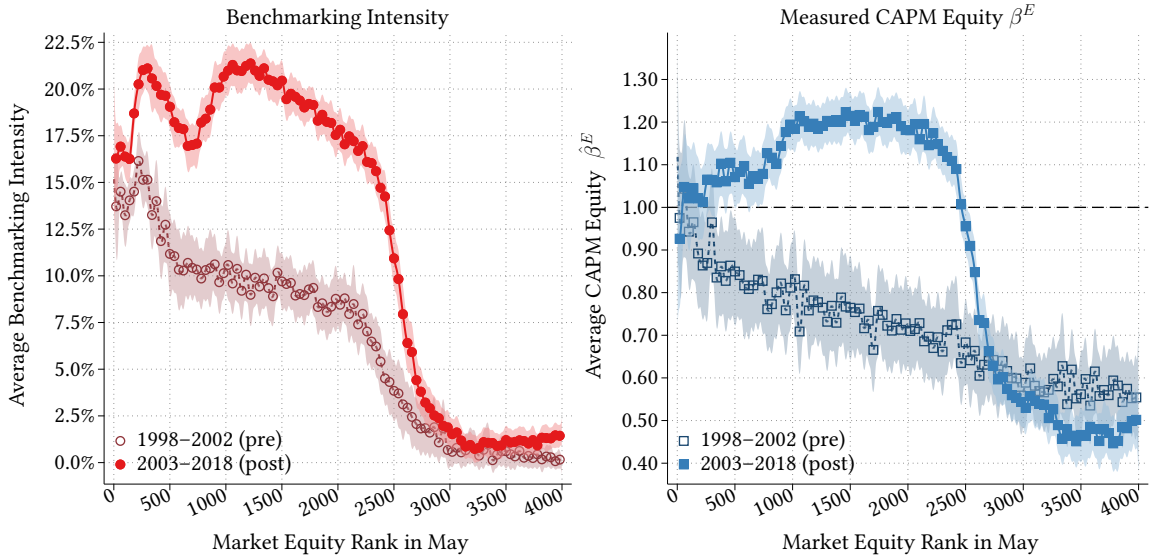
Figure B1: Scholes-William Non-Synchronous Rolling-Window CAPM $\widehat{\beta}$ Estimator



Notes: This figure shows binned scatter plots of rolling-window CAPM $\widehat{\beta}$s using the Scholes-Williams (1977) estimator against the market equity rank in May. The Scholes-Williams estimator adjusts the traditional β calculation for nonsynchronous trading by incorporating lagged and leading market returns alongside current returns. This adjustments account for potential biases in measuring systematic risk due to infrequent or nonsynchronous trading. Daily rolling window estimates use 252 trading days. Shaded areas are 90% confidence bands based on standard errors clustered at the stock and year-month level.

Figure B2: Binned Scatter Plots of BMI and CAPM $\widehat{\beta}$ against Market Equity Rank (bin width: 40)

(a) Benchmarking Intensity and CAPM Equity $\widehat{\beta}^E$ versus Market Equity Rank in May



Notes: This figure shows binned scatter plots of stocks' monthly BMI and CAPM equity $\widehat{\beta}^E$ estimates vs. their market equity rank in May. The estimation uses cross-sectional variation by absorbing year-month fixed effects to identify the conditional means across market equity ranks. Each bin shows the equal-weighted average of approximately 40 stocks. Outlined bins show estimates from 1998 to 2002, while filled bins show estimates from 2003 to 2018. Shaded areas are 95% confidence bands based on standard errors clustered at the stock and year-month level.

(b) Differences in Benchmarking Intensity and CAPM Equity β^E Between Pre and Post

